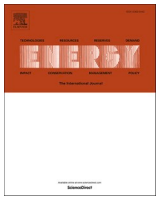
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Comprehensive performance assessment of photovoltaic/thermal system using MWCNT/water nanofluid and novel finned multi-block nano-enhanced phase change material-based thermal collector: Energy, exergy, economic, and environmental (4E) perspectives

Praveen Kumar Tyagi, Rajan Kumar *

Department of Mechanical Engineering, Dr B R Ambedkar National Institute of Technology, Jalandhar, Punjab, 144008, India

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ABSTRACT

An innovative multi-block finned nano-enhanced phase change material (NePCM)-based photovoltaic/thermal (PV/T) system with thermal regulations through deionized water (DIW) and nanofluid at 0.25, 0.5, and 0.75 wt% is designed, fabricated, and tested in this study. The main goals are to achieve consistent heat transfer distribution, mitigate temperature stratification within each thermal energy storage block, and prevent material shrinkage at the bottom. The energy and exergy performance of the proposed PV/T-NePCM is investigated using indices such as power output, efficiency, exergy destruction, entropy generation, and sustainability index. Furthermore, environmental metrics such as CO_2 emission, carbon credit gained, and net CO_2 mitigation, as well as economic performance parameters such as energy payback time and energy cost are analyzed. The proposed system uses nanofluid as the heat transfer fluid at 1.5 l/min and 0.75 wt% NePCM, resulting in 9.26 % and 2.93 % higher energy and exergy efficiency than the DIW-based system. The economic evaluation shows that the PV/T-NePCM system has a cost of energy of 5.24 INR/kWh (0.063 USD/kWh), while the conventional PV module costs 5.64 INR/kWh (0.069 USD/kWh) with an 8 % interest rate and a 20-year lifespan. For environmental sustainability, the nanofluid-cooled PV/T-NePCM system has 637.28 kWh of embodied energy.

1. Introduction

The recent surge in industrial expansion and urban development has undeniably increased the energy demand, increasing dependency on conventional fossil fuels. However, this growing reliance presents critical issues, particularly in preventing global warming and lowering CO_2 emissions into the environment [1]. There is a rising need to shift to alternate energy sources to solve these critical concerns, indicating a paradigm shift away from the conventional reliance on fossil fuels [2,3]. Adopting renewable and sustainable energy solutions is critical to addressing rising energy demands and guaranteeing a more environmentally conscious and sustainable future [4–6]. Solar energy, an important renewable energy source, can be harvested in various ways. One way involves converting solar energy into heat using a solar thermal energy system. Another alternative is using photovoltaic (PV) systems, which convert solar power into electrical power [7–9].

Additionally, there is the photovoltaic/thermal (PV/T) system, capable of simultaneously generating electricity and heat from solar energy [10,11]. Effective thermal management is vital for optimizing system efficiency and extending its lifespan by keeping operating temperatures within an optimal range [12,13]. Proper thermal control helps prevent overheating, minimize energy losses, and improve overall performance and reliability, especially in renewable energy systems. The PV/T systems use three cooling solutions to regulate the thermal performance of PV modules: active, passive, and hybrid cooling. To extract heat from the PV module, active cooling systems require an external power supply [14]. These systems can be classified into two primary groups based on the working fluid utilized: those employing air and those employing liquids such as water, oil, and ethylene glycol [15]. However, water is frequently chosen as the preferred working fluid in PV/T systems due to its superior thermal capacity and improved heat transfer characteristics, making it more effective for thermal regulation than air-based systems.

In the realm of modern technologies, the adoption of nanoadditives and nanofluids, distinguished by their notably enhanced thermal

* Corresponding author.
E-mail addresses: rajan.rana9008@gmail.com, rajank@nitj.ac.in (R. Kumar).

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Nomenclature

Abbreviations

AFC	Annual first cost
ARC	Annual running cost
AMC	Annual maintenance cost
ASV	Annual salvage value
CC	Capital cost
CNTs	Carbon nanotubes
CRF	Capital recovery factor
DIW	Deionized water
DSC	Differential scanning calorimetry
EPBT	Energy payback time
FESEM	Field Emission Scanning Electron Microscopy
FT-IR	Fourier transform infrared
MWCNT	Multi-walled carbon nanotubes
NePCM	Nano-enhanced Phase change material
PCM	Phase change material
PV	Photovoltaic
PV/T	Photovoltaic/thermal
PV/T-NePCM	Photovoltaic/thermal system integrated with NePCM
PV/T-PCM	Photovoltaic/thermal system integrated with PCM
S	Salvage value
SDBS	Sodium dodecyl benzene sulfonate
SFF	Shrinkage fund factor
TAC	Total annual cost
TGA	Thermogravimetric analysis

Symbols

E_{el}	Electrical power output (W)
T_m	Phase change temperature (°C)
Q_{th}	Thermal power output (W)
E_{ov}	Overall power output (W)
E_{Sun}	Sun energy (W)
Ex_{Sun}	Sun exergy (W)
η_{el}	Electrical energy efficiency (%)
η_{th}	Thermal energy efficiency (%)
η_{ov}	Overall energy efficiency (%)
G	Incident solar irradiation (W/m^2)
T_{amb}	Ambient temperature (°C)
Ψ_s	Sun factor
T_{Sun}	Sun equivalent temperature (°C)
Ex_{el}	Electrical exergy output (W)
E_{pump}	Electrical pumping power (W)
ΔP_{drop}	Pressure drops (Pa)
Ex_{th}	Thermal exergy output (W)
T_m	Mean fluid temperature (°C)
Ex_{ov}	Overall exergy power (W)
Ex_{des}	Exergy destruction (W)
S_{gen}	Entropy generation (W/K)
$\eta_{ex_{el}}$	Electrical exergy efficiency (%)
$\eta_{ex_{th}}$	Thermal exergy efficiency (%)
$\eta_{ex_{ov}}$	Overall exergy efficiency (%)
N_S	Annual number of days
N	Life span
R	Interest rate
k_T	Thermal conductivity (W/m·K)
E_{emb}	Embodied energy (kWh)

conductivity (k_T), stands as a contemporary trend [16–18]. This approach effectively mitigates or eliminates the adverse effects of elevated temperatures on PV modules, achieved through meticulous thermal regulation [19]. Attia et al. [20] conducted a numerical analysis to evaluate the thermal performance of a PV/T system using air and water. The system was tested with two different flow configurations: a typical single straight zigzag plated flow and a three zigzag plated flow configuration. The second option yields the most optimal outcomes. The thermal energy efficiency (η_{th}) of the water and air-based systems increased by 7.23 % and 15.75 %, respectively, compared to the traditional PV/T system. Alkhalidi et al. [21] conducted an experimental study to assess the performance of PV/T systems of both open-loop and closed-loop types at various tilt angles. The study found that the open loop system had a superior overall energy efficiency (η_{ov}) compared to the closed loop system, with a 31.04 % advantage attributed to its better thermal efficiency.

In a recent study by Alktranee et al. [22], the investigators conducted an experimental investigation on the performance of a ZrO_2/deionized water (DIW) nanofluid in a PV/T system. The results showed a significant reduction in the temperature of the PV module, with a decrease of 10.2 °C perceived at a concentration of 0.0275 vol%. Venkatesh et al. [23] investigated the performance of graphene-based PV/T systems at various flow rates and concentrations (0–3 vol%). The investigated system's electrical energy output (E_{el}) was determined to be 10 W and 3 vol%. Praveenkumar et al. [24] assessed the performance of a PV/T system that uses water as a coolant, as well as a PV/T system that utilizes an Al_2O_3/water nanofluid as a coolant in conjunction with a thermoelectric generator. The latter showed reduced net CO_2 emissions by 2.07 tonnes annually, resulting in a carbon credit worth 51.54 United States Doller (USD) annually. Another study on a PV/T system coupled with a thermoelectric generator was numerically investigated by Sheikholeslami et al. [25]. Gharapetian et al. [26] investigated a PV/T system integrated with a thermoelectric generator, utilizing water, CuO/water, and Fe-CuO/water as working fluids. Their findings revealed that the Fe-CuO/water exhibited the lowest upper surface temperature of the PV module, achieving reductions of 8.8 % and 13 % compared to the CuO/water and water coolants, respectively. The study employed MgO/water nanofluid as a spectral filter, leading to a CO_2 emission decrease that was twice as significant as that achieved by the conventional module. Sheikholeslami and Khalili [27] conducted a numerical investigation on the performance of a PV/T system utilizing a graphene nanoplatelets/water nanofluid. They found that this coolant had an optimal solution for enhancing the system's performance. In their study, Bhat and Qayoum [28] investigated the efficiency of a PV/T system using two types of pipes: a regular straight pipe and a hybrid pipe. Various types of coolants were employed, such as ethylene glycol and mono and hybrid nanofluid made of MWCNT and CuO nanomaterial. The binary nanofluid exhibited a superior η_{ov} of 9.01 % higher in comparison to ethylene glycol. Abdalla and Shahsavar [29] examined the performance of a PV/T system utilizing a ternary nanofluid composed of TiO_2-Fe_2O_3-Graphene oxide/water. The study focused on flow rates ranging from 20 to 80 kg/h and concentrations varying from 0 to 1 vol%. The system's overall exergy efficiency ($\eta_{ex_{ov}}$) was observed to be 15.36 % when operating at the ideal flow rate of 80 kg/h and 1 vol % concentration. Ghalambaz et al. [30] employed the finite element method to numerically investigate the natural convection movement of a hybrid Ag-MgO/water nanofluid within a square cavity. The findings indicate that the heat transfer rate increases as the nanofluid concentration increases until it reaches the conduction-dominant region.

Passive cooling extracts heat from PV modules without using an external power source. Integrating phase change material (PCM) into PV/T systems (PV/T-PCM) can serve as an effective passive method for regulating the temperature of PV modules and enhancing the overall efficiency of PV/T systems [31]. In their study, Kumar et al. [32]

[examined the effectiveness of PCM-filled PV modules with and without external fins in the Indian environment. They found that PV modules with external fins exhibited an average electrical energy efficiency (η_{el}) of 6.59 % greater than that of conventional module. Several studies have been carried out to enhance the power output of the PV module by including passive cooling methods, such as fins or PCM, or a combination of both, in different operating conditions [33–35]. A recent study by Jamil et al. [36] found that integrating carbon black and SiO_2-enhanced PCM at 0.25 and 0.5 wt% in a PV module improved its performance. The integrated system with carbon black achieved an η_{el} of 12.07 %, while the system with SiO_2 achieved an efficiency of 12 %.

Hybrid cooling refers to the integration of both active and passive cooling methods. Kazemian et al. [37] experimentally investigated the performance of war and ethylene glycol-based PV/T-PCM systems. The $\eta_{ex_{ov}}$ for the investigated fluids, conventional water, ethylene glycol (50 %), and ethylene glycol (100 %) were found to be 14.37 %, 14.42 %, and 15.13 %, respectively. Different studies have been conducted to improve the overall performance of water-based PV/T-PCM systems [38–40]. Some studies have also been carried out to enhance the overall performance of nanofluid-based PV/T-PCM systems [41,42]. A recent study is experimentally conducted by Prakash et al. [43] to examine the effectiveness of water-cooled PV/T and PV/T-PCM systems regarding energy and exergy considerations. The study utilized a butterfly-shaped flow channel configuration and hydrated salt as a PCM. Results indicated an $\eta_{ex_{ov}}$ of 13.9 and 23.54 % for the investigated systems, respectively, compared to conventional PV modules. Rao et al. [44] also experimentally investigated the energy efficiency of a PV/T-PCM system with a two-sided serpentine flow channel configuration. They used lauric acid having melting temperature (T_m) = 44–46 °C and tested it at different water flow rates ranging from 0.0085 to 0.067 kg/s. The PCM is integrated into the system using Al bags. The system's efficiency increased by 23 % when the flow rate was optimized at 0.033 kg/s, compared to 0.0085 kg/s. Asefi et al. [45] conducted a numerical evaluation of the techno-economic feasibility using MATLAB. Also, they performed experimental tests on a PV/T system that was combined with a porous PCM (lauric acid). This technique successfully decreased the annual CO_2 emission by 30 tonnes. In their study, Sehrawat et al. [46] investigated the enviro-economic performance of a sustainable domestic drying system combined with a bifacial PV/T-PCM system. The system acquired Carbon credits worth 498,514 USD. Gharapetian et al. [47] performed a numerical analysis to evaluate the performance of a PV/T system that included an Al_2O_3/water nanofluid and three distinct PCMs with porosities of 85, 90, and 95 %. The $C_{18}H_{30}$ PCM with 85 % porosity exhibited the most favorable outcomes. The nanoencapsulation of PCMs provides notable benefits in thermal performance and stability [48,49] but also brings several drawbacks. These include higher costs, complex manufacturing processes, potential reductions in latent heat storage per unit mass, and challenges with thermal resistance. These considerations are crucial when deciding between nanoencapsulated PCMs and base PCMs for specific applications. Hamada et al. [50] numerically investigated the thermodynamic and optical performance of a double pass PV/T system associated with nano-encapsulated PCM composed of silica shell and core made of paraffin wax. This system significantly decreased the temperature of the upper surface of the PV module. The thermal performance of flat-plate solar collectors can be effectively enhanced by increasing the collector dimensions. Additionally, large-scale flat-plate solar collectors outperform conventional flat-plate solar collectors when used in parallel [51]. Air collectors with superior freezing resistance are frequently utilized for solar collector fields in remote areas to lower maintenance costs [52]. Mo et al. [53] proposed a dual-objective optimization model for balancing economic and heating performance. It is suggested that lowering the cost of solar collectors and heat storage tanks might help in the development of more competitive solar heating system designs. Zhang et al. [54] utilized NSGA-II to apply optimized techniques to examine the techno-economic performance of concentrated solar heating systems integrated with water-to-water heat pumps. The TRNSYS model, developed by Ref. [55], focuses on flat plate-based thermal collectors. They analyzed and optimized the energy and economic aspects of hot water systems.

Temperatures in northern India, especially in the experimental site Jalandhar, Punjab, intensified considerably during the summer, raising the temperature of the PV cell. These high temperatures adversely affect the performance of PV cells, resulting in a decrease in both conversion efficiency and the lifespan of the PV module. Therefore, it is imperative to lower the cell's temperature using effective cooling methods to increase the power generated by the cell. Incorporating PCM into PV/T systems is a viable solution for effectively managing the thermal management of PV modules and improving the overall performance of PV/T systems. Different methods, ranging from Al foil pouches to single metal and acrylic glass containers, are used to integrate PCMs into PV/T systems to manage their thermal regulation. However, proper container design is crucial, especially for sealed containers containing significant amounts of PCM, to account for variations in PCM content due to density differences between solid and liquid states. Inadequate design can lead to container damage and substantial system losses [56]. Stringent safety measures must be prioritized in container design and maintenance to prevent accidents and protect personnel [57]. Incorporating a large thermal energy storage container into a solar panel presents difficulties.

Additionally, base PCMs exhibit lower k_T [58] and may experience performance degradation over time, but nanomaterials enhance heat transfer rates, boosting total recoverable energy. However, research gaps persist in concurrently integrating both active and passive cooling methods in PV/T systems with nanoadditive integration in both heat transfer fluid and PCMs. Further research is needed to understand the effective integration of multi-block finned containers with PV modules for analyzing energy, exergy, economic, and environmental (4E) prospects. Thus, the present design incorporates a multi-block finned absorber plate-type thermal collector to ensure even heat distribution within each thermal energy storage block, preventing material shrinkage at the bottom. Moreover, it improves the material's k_T by including nanomaterials and addresses PCM leakage issues. This design system adopts a hybrid cooling strategy, combining two techniques: the use of multi-walled carbon nanotubes (MWCNT)/water nanofluid employing as an active cooling technique and MWCNT-enhanced PCM (NePCM), utilized as a passive cooling technique. Therefore, this study investigates the feasibility and benefits of NePCM and nanofluid

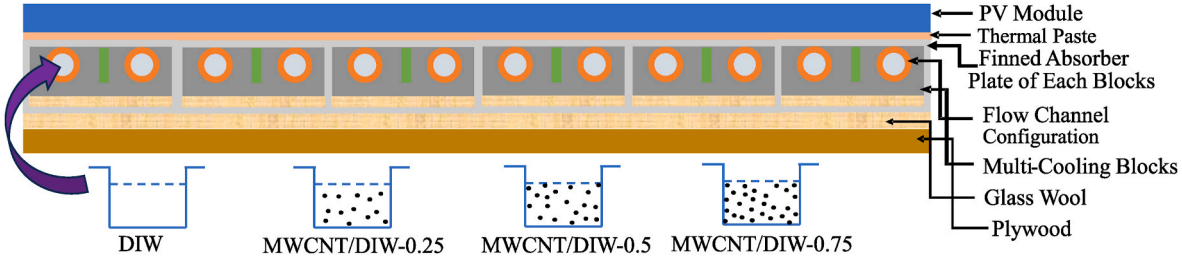


Fig. 1. Layout of a side view of innovative system design.

Table 1
Specifications and dimensions of the proposed thermal collector.

Particular	Specification/ dimension	Particular	Specification/ dimension
Thermal collector type	Sheet and Tube	Collector material	Al
Tube material for coolant flow	Cu	Number of cooling blocks	6 Nos.
Dimension of thermal regulating block -I	100 ×295×25 mm	Dimension of thermal regulating block -II	100×465×25 mm
Inner diameter tube	10 mm	Outer diameter tube	12 mm
Utilized fin dimension-Type I	290×22 mm	Utilized fin dimension- Type II	460×22 mm
Nanoadditive utilized	MWCNT	Purity of nanoadditive	>98 %
Dimension of nanoadditive -outer diameter	30–40 nm	Specific surface area of nanoadditive	330 m^2/g
Appearance/color of nanotubes	Black Powder	Bulk density of nanotubes	0.06–0.09 g/cm^3
$k_{T,MWCNT}$	2586 W/m·K	Length of nanoadditive	5–10 μm
Thermal energy storage material	Paraffin wax as PCM	Heat storage capacity of PCM	195 kJ/kg
PCM phase change temperature	53.4 °C	$k_{T,PCM}$	0.19 W/m•K
Insulation type-I	Glass wool	Insulation type-II	Plywood
$k_{T,Insulation-I}$	0.032 W/m•K	$k_{T,Insulation-II}$	0.1154 W/m•K
Thickness of insulation layer-I and II	20 mm	Dimension of insulation layer -I and II	540 × 665 mm

integration in PV/T systems under such conditions. It offers valuable insights into enhancing sustainability and efficiency, addressing the challenges posed by significant temperature variations.

This study will be initiated by elucidating the design description and specification associated with the innovative design of the nanofluid-based PV/T-NePCM system. Further details of the experimental setup and its procedures, as well as an overview of the properties and characteristics of the employed nanofluid and NePCM, are also presented. Additionally, it delves into the mathematical equations of performance evaluation indices and addresses the uncertainty associated with the utilized instruments in the analysis. The study thoroughly examines and discusses outcomes related to ambient conditions, PV module temperature, and perspectives on energy, exergy, economic, and environmental aspects.

2. Methodology

2.1. Innovative PV/T-NePCM system design description and specifications

This design system showcases a thermal collector with a multi-block finned absorber plate type design. Each NePCM heat sink block is positioned directly below the PV module in this proposed PV/T-NePCM system. Fig. 1 displays the layout of a side view of an innovative design with multi-block finned thermal regulatory cooling blocks.

A polycrystalline PV module with a maximum capacity of 50 W is used in the experimental setup. To enhance the efficiency of the cooling blocks, the thermally conductive paste is utilized on the bottom surface of the PV and the upper surface of the thermal collector. This application aims to minimize the thermal resistance between the contacting surfaces. Table 1 provides detailed information about the specifications and dimensions of the thermal collector.

2.2. Experimental facility and testing procedures

Once the conceptualization and fabrication stages have been completed, the various subsystems can be integrated to make the PV/T-NePCM. Fig. 2(a–f) outlines the assembly procedures for the proposed system. In (a), the initial stage involves forming each block container from aluminum sheets. Image (b) shows the final stage of the block container, which has a circular Cu pipe for fluid circulation and a fin made of Al for heat dissipation within each block. In (c), all six finned containers are installed beneath the PV module using thermal conductive paste. Images (d) and (e) depict layer 1 and layer 2 insulation, made of glass wool and plywood, respectively. Finally, (f) shows the complete assembly of a multi-block absorber plate-based thermal collector. The actual experimental front and rear views are presented in Fig. 3.

To minimize potential errors caused by unanticipated environmental changes such as overcast weather or fluctuating wind speeds, investigations were undertaken during the relatively stable meteorological circumstances of September and October 2023 in the actual climate conditions of Jalandhar, Punjab, India. The coolant temperature at inlet and out and NePCM temperature within each finned cooling container are measured using PT-100 sensors. Additionally, the upper surface temperatures of the conventional PV module and the proposed system are measured using K-type thermocouples. A thermal imaging infrared camera (Testo-885) is employed to compare the temperature distribution of the tested systems at different time intervals, and the captured images are analyzed using IRSoft software. The coolant is stored in a tank and continuously circulated through the flow configuration using a direct current (DC) pump, ensuring a constant mass flow rate. Table 2 describes the different coolants used in the current investigation.

The selected coolant volumetric flow rates (0.5 l/min, 1 l/min, and 1.5 l/min) are associated with varying levels of valve aperture. A heat exchanger is employed to establish a closed-loop circulation system for the coolant, enabling the coolant to cool after it has been heated in the thermal collector. The primary aim was to evaluate the performance of PV/T-NePCM systems in terms of energy, exergy, environmental impact, and economic feasibility, utilizing both DIW and nanofluid. Therefore, this study highlights the following works.

- Selecting the optimal nanomaterial doping loading condition (0.25, 0.5, 0.75, 1, 1.5, and 2 wt%) for NePCM
- Using DIW at different volumetric flow rates of 0.5, 1, and 1.5 l/min in the PV/T-PCM system and selecting the optimal flow rate

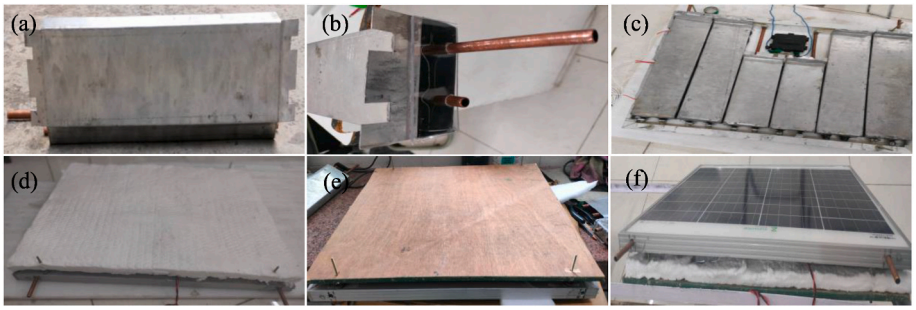
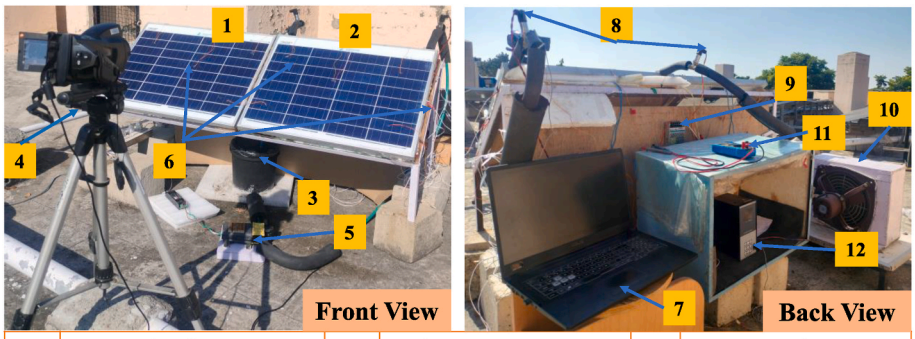


Fig. 2. Fabrication steps of the proposed PV/T-NePCM system.



1.	Conventional PV	2.	PV/T-NePCM	3.	Storage Tank
4.	Thermal Imaging Camera	5.	DC Pump	6.	Thermocouples
7.	Laptop	8.	Pressure Transducer	9.	AC to DC Converter
10.	Radiator	11.	Multimeter	12.	Data Logger

Fig. 3. Front and back view of the actual experimental setup.

Table 2
Description of the different coolants employed in the current investigation.

Abbreviation	Description of proposed coolant
DIW	Deionized water (DIW)
MWCNT/DIW-0.25	Nanofluid produced through homogeneously dispersing MWCNT at 0.25 wt% concentration in DIW
MWCNT/DIW-0.5	Nanofluid produced through homogeneously dispersing MWCNT at 0.5 wt% concentration in DIW
MWCNT/DIW-0.75	Nanofluid produced through homogeneously dispersing MWCNT at 0.75 wt% concentration in DIW

- Under the chosen NePCM loading condition and optimal flow rate, the investigated PV/T-NePCM system undergoes performance analysis using different heat transfer fluids: DIW, MWCNT/DIW-0.25, MWCNT/DIW-0.5, and MWCNT/DIW-0.75
- Evaluate the proposed system's energy and exergy performance using indices such as power output, efficiency, exergy destruction, entropy generation, and sustainability index. Furthermore, environmental metrics such as CO_2 emission, carbon credit gained, and net CO_2 mitigation, as well as economic performance parameters such as energy payback time and energy cost, are evaluated

2.3. Preparation, characterization, and stability of MWCNT/DIW nanofluid

MWCNT is chosen as a nanomaterial additive for the current study due to its exceptional thermal characteristics. The nanoadditive is supplied by the Nano Research Lab in Jamshedpur, Jharkhand, India. Nanofluid has the potential for commercial applicability when the benefit of increased heat transfer outweighs the penalty of increasing pumping power. Therefore, a low concentration of nanomaterials is utilized in the present study. The MWCNT/DIW nanofluid is synthesized through a two-step process. However, the hydrophilic characteristics of MWCNT create difficulties in achieving dispersion because of the firm van der Waals forces between carbon surfaces and their high aspect ratio. Nevertheless, it is achievable to make a stable nanofluid by adding appropriate surfactants to a suspension and carefully adjusting agitation parameters such as stirrer speed, ultrasonication time, and methods.

The process for preparing the utilized nanofluid is depicted in Fig. 4. Sodium dodecyl benzene sulfonate (SDBS) surfactant is added to the base fluid, which is DIW, to reduce the surface tension. The MWCNTs were subsequently stirred in the surfactant-based mixture using a Magnetic Stirrer for 30 min, provided by Labman Scientific Instruments Pvt. Ltd, India. The combination is subjected to sonication using a Probe

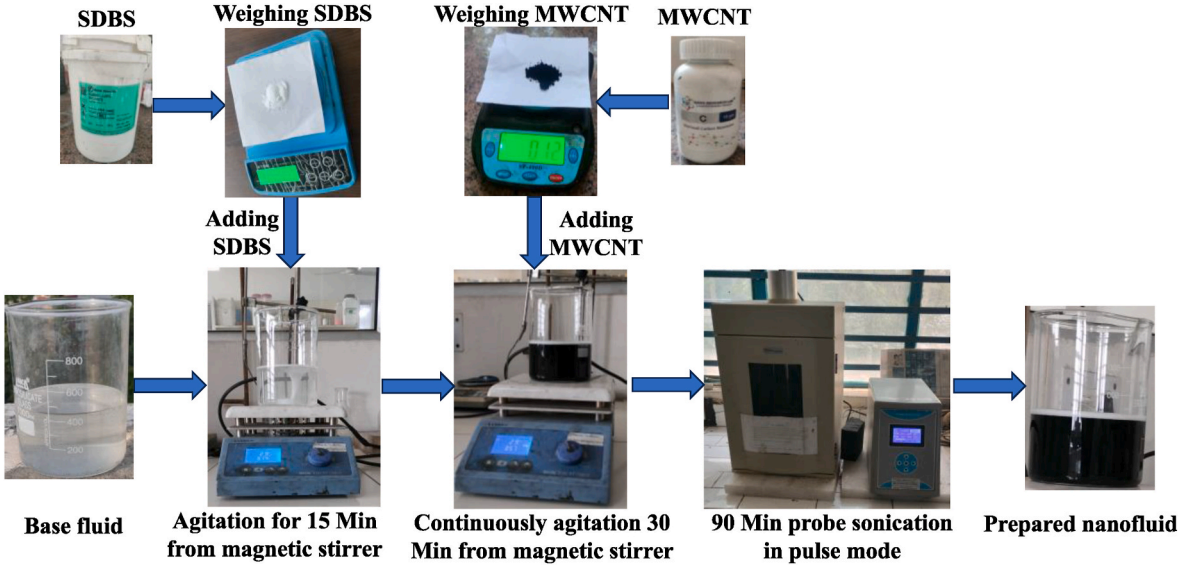


Fig. 4. Preparation procedure of utilized nanofluid.

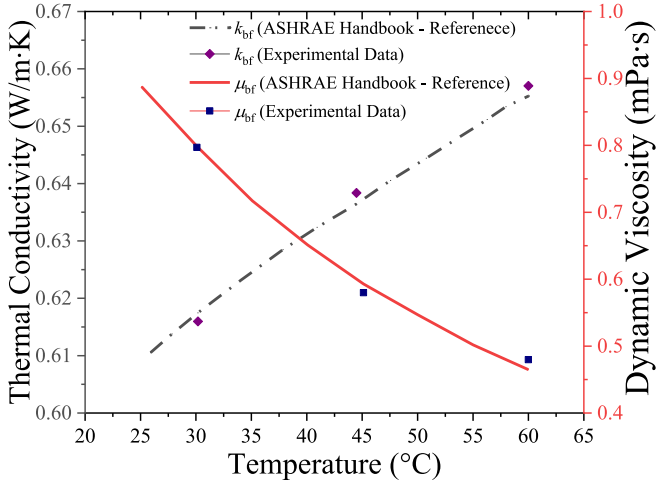


Fig. 5. Calibration test of k_T and dynamic viscosity data.

Sonicator in pulse mode (2 s ON and 2 s OFF) for 90 min to achieve a uniform mixture. Bandi Technology, New Delhi, India, provides this sonicator.

Prolonging the ultrasonication duration leads to a progressive reduction in the dynamic viscosity of the samples. This effect occurs because longer sonication breaks down MWCNT particles into smaller sizes, de-clustering MWCNT bundles. However, extending the ultrasonication beyond 90 min results in an increase in viscosity. The optimal ultrasonication time of 90 min achieves the minimum viscosity due to a more uniform distribution of MWCNTs.

The physical properties of the coolant, including its density, heat capacity, and viscosity, as well as k_T, significantly impact the performance of the solar collector system [59]. The k_T is measured using a handheld Thermal Conductivity Analyzer (METRE-TEMPOS 1546) equipped with different sensors. To precisely measure the k_T of DIW and nanofluid samples, a KS-3 sensor with a 1.3 mm diameter and 60 mm length is utilized, which offers an accuracy of ±0.01 %. It can measure k_T within 0.02–2 W/m·K range. A calibration test is conducted before experimentation to confirm the accuracy and reliability of the device with glycerol provided by the manufacturer and pure water. The test results are shown in Fig. 5, showing measured k_T data of pure water under a temperature range of 30–60 °C and comparing the value with a reference available in the ASHRAE Handbook [60].

The experiment showed good agreement between the measured data and the reference data, confirming the accuracy of the measurements. A capillary or Ostwald viscometer is used to measure the viscosity of the prepared samples. The samples are kept at a consistent temperature in an oven bath provided by DAIHAN Scientific. Before conducting the test, a calibration is performed by measuring the viscosity of pure water and comparing the results with reference data from the ASHRAE Handbook. This comparison, shown in Fig. 5, demonstrates a strong correlation and confirms the accuracy of the measured data.

Each experiment is conducted three times to assess the repeatability of the results, and the average values are recorded. The behavior of the samples is examined at three temperatures: 30 °C, 45 °C, and 60 °C. Fig. 6(a) illustrates the variations in k_T with temperature for different tested concentrations. It is evident that k_T increases with rising temperature. This trend is primarily due to the increased Brownian motion of particles at higher temperatures, which enhances k_T. The increase in kinetic energy due to the temperature rise, along with a higher rate of molecular collisions and reduced thermal resistance between nanoadditives and the base fluid at elevated temperatures, may also contribute to the increased k_T observed in the tested nanofluids.

Additionally, incorporating MWCNT nanoadditives into the base fluid improves the k_T of the nanofluid across all studied concentrations. The most significant enhancement was observed at a concentration of 0.75 % and a temperature of 60 °C. This substantial increase is attributed to the high k_T of MWCNT nanoadditives, which boosts the overall k_T of the nanofluid. Fig. 6(b) shows that the viscosity consistently decreases with rising temperatures, regardless of solid concentrations. This reduction is primarily due to the weakening of van der Waals forces (inter-particle adhesion forces) as temperature increases.

The heat capacity of nanofluids is influenced by the concentration and type of nanoadditives, the type of base fluid, and the system temperature. It decreases as the concentration of nanoadditives in the base fluid increases because nanoadditives typically have a lower heat capacity than the base fluid. The heat capacity of tested nanofluids ranges

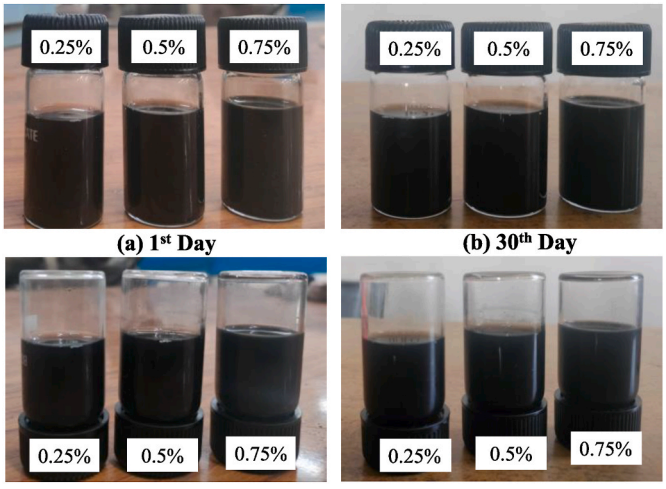


Fig. 7. Nanofluid stability by visual inspection (a) fresh samples and (b) after 1 month.

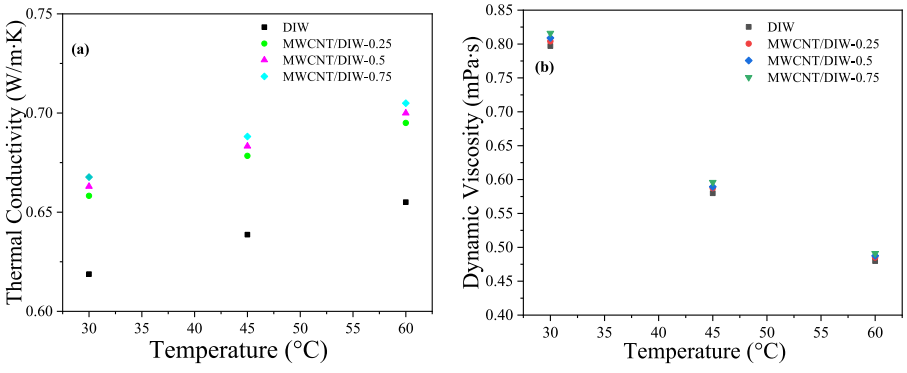


Fig. 6. (a) k_T and (b) dynamic viscosity variation of base fluid and nanofluid at different temperatures and loading conditions.

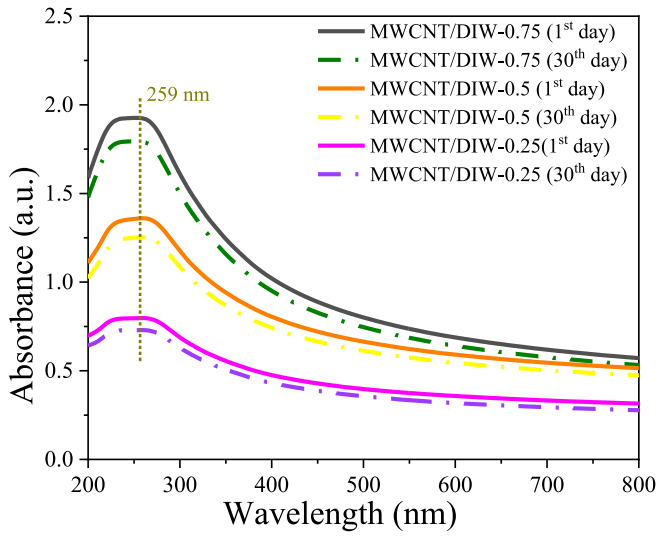


Fig. 8. Absorbance spectra vs wavelength plot of MWCNT/DIW nanofluids.

from 4172 to 4165.11 J/kg·K for concentrations of 0.25–0.75 wt% at 30 °C. Furthermore, raising the temperature tends to boost the heat capacity. MWCNT/DIW-0.75 has a heat capacity of 4165.12–4170.74 J/kg·K at temperatures ranging from 30 to 60 °C.

The stability of MWCNT/DIW nanofluid is determined first by visual inspection by taking pictures on the fresh sample day and the 30th day after sample preparation, as shown in Fig. 7. This reveals that no observable deposition is seen in all the tested concentrations of 0.25, 0.5, and 0.75 wt%, even after the 30th day of nanofluid preparation. Varying concentrations of nanofluid can exhibit varying levels of stability. As the concentration of nanoadditive in the base fluid increases, the likelihood of aggregation and sedimentation also increases.

Additionally, a Shimadzu-2600 Ultraviolet–Visible (UV–Vis) spectrometer is used to quantitatively assess the dispersion stability of the nanofluids. It is accomplished by analyzing the absorbance levels within the 200–800 nm wavelength range, as depicted in Fig. 8. The absorbance spectra acquired on the 1st and 30th days after sample preparation, utilizing a solitary concentration of nanotubes, demonstrate remarkably similar overall patterns concerning shape and absorption peak values. The light absorption intensity is wider immediately after sample preparation compared to 30th days after preparation. However, the observed absorbance levels for the tested nanofluids peak within the same wavelength range of 200–300 nm. Therefore, MWCNT/DIW nanofluid at different concentrations produced using the described procedure demonstrates stability for over a month, as confirmed by this stability analysis.

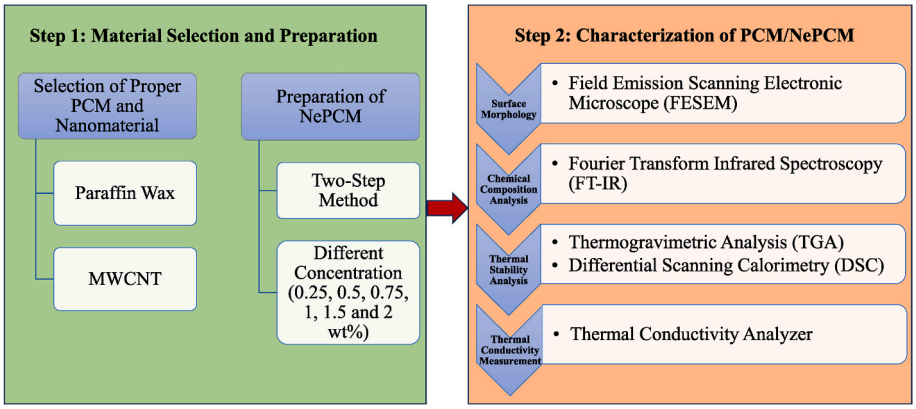


Fig. 9. Framework of selection, preparation, and characterization for used NePCM.

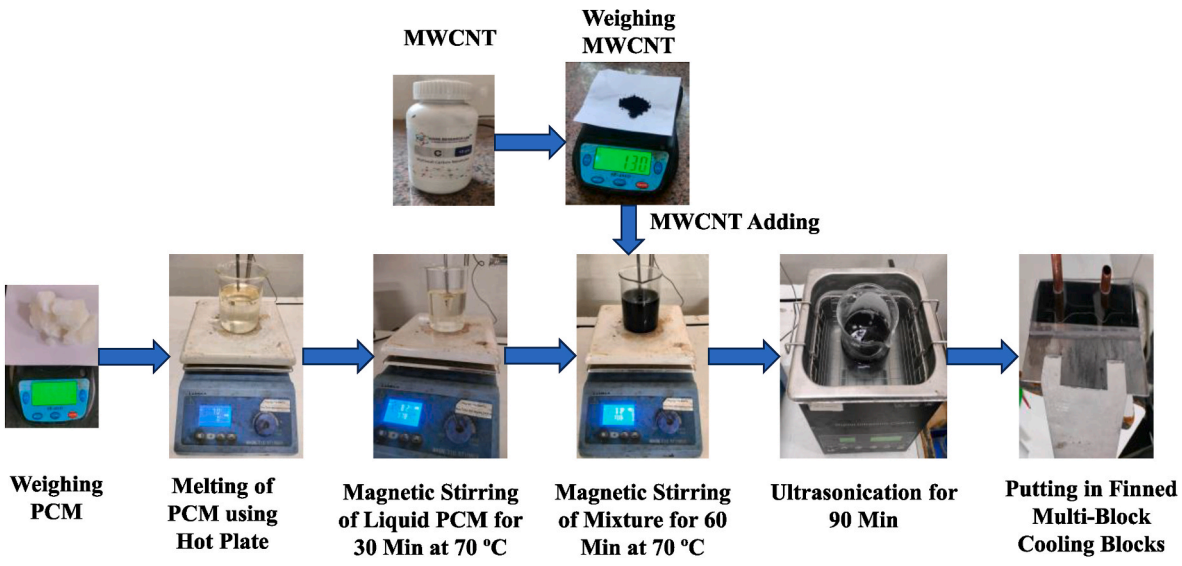


Fig. 10. Preparation process of used NePCM.

2.4. Selection, preparation, and characterization of NePCM

Fig. 9 provides a comprehensive outline for the selection, production, and characterization of NePCM. The present study uses paraffin wax as PCM based on the prevailing environmental circumstances. It is readily accessible and possesses improved safety features and exceptional degrading properties within a specific temperature range. However, the k_T of paraffin wax is lesser, which can be enhanced by doping with nanomaterial. Therefore, MWCNT is selected as a nanomaterial due to its higher k_T and lower density compared to metallic and non-metallic nanomaterials. The selected loading conditions of nanomaterial doping were 0.25, 0.5, 0.75, 1, 1.5 and 2 wt%.

The NePCM is produced using a two-step methodology. This method is economically efficient and may be readily implemented due to the availability of nanomaterials produced by numerous global firms. Fig. 10 illustrates the procedure for preparing NePCM. The digital weighing apparatus is employed to measure the weight of MWCNT and solid paraffin wax. The Magnetic Stirrer is utilized to agitate a mixture of PCM and nanomaterial at a speed of 700 rpm for approximately 60 min while maintaining a temperature of 70 °C.

This process consequences in the formation of a uniform and consistent blend. The manufacturing specifications of the stirrer have already been provided in the nanofluid preparation section. The combination is then exposed to sonication using a digital ultrasonicator from Labgear International at a frequency of 33 kHz for 90 min to ensure a consistent mixture. Next, place the prepped NePCM onto each finned cooling block.

The characterization of prepared samples of NePCM at different concentrations and base PCM is governed by various equipment. The surface topography of PCM/NePCM samples is analyzed using Field Emission Scanning Electron Microscopy (FESEM) with a ZEISS (Sigma 500 VP) model. To prevent the production of oxides caused by high-temperature scanning electrons, the images were captured in an atmosphere of nitrogen. The chemical stability of the NePCM samples produced is examined using Fourier transform infrared (FT-IR) spectroscopy, explicitly utilizing the Agilent Cary 630 equipment. This work is crucial for understanding the chemical interactions between nanomaterials and the basic PCM. Thermogravimetric analysis (TGA) is employed to assess the thermal stability of the prepared samples. This analysis is performed using a Thermal Gravimetric Analyzer, namely the Perkin Elmer-STA 600 model. The differential scanning calorimetry (DSC) analysis is employed to obtain the melting temperature of prepared samples of PCM/NePCM. For measuring the k_T of PCM/NePCM samples, the Thermal Conductivity Analyzer is equipped with a TR-3 sensor having a 2.4 mm diameter and 100 mm length.

2.5. Performance evaluation indices

The primary objective of this section is to evaluate performance measures to predict the PV/T-NePCM system performance in terms of energy, exergy, economics, and the environment.

The energy balance equation is expressed in the general form as follows:

$$\sum E_{in} = \sum E_{out} + \sum E_{loss} \tag{1}$$

$$E_{Sun} + E_{mass,in} = E_{mass,out} + E_{el} + \sum E_{loss} \tag{2}$$

Where E_{Sun} is the sun's energy, $\sum E_{loss}$ is the summation of lost energy due to convection, radiation, and friction. E_{el} is the electrical power output. E_{in} is input energy rate and E_{out} is output energy rate.

The electrical power output (E_{el}) of the tested systems can be evaluated by:

$$E_{el} = V_{oc}\, I_{sc}\, FF \tag{3}$$

Where I_{sc}, FF, and V_{oc} are the short circuit current, filling factor, and open circuit voltage, respectively. FF reflects the highest power conversion output of a PV module.

The thermal power output (Q_{th}) can be evaluated by Equation (4):

$$Q_{th} = E_{mass,out} - E_{mass,in} = m_{wf} c_{p,wf} \left(T_{wf_2} - T_{wf_1}\right) \tag{4}$$

Where $m_{wf}, c_{p,wf}$, T_{wf_2} and T_{wf_1} represent the coolant flow rate, heat capacity at constant pressure, and working fluid outlet and inlet temperature.

Electrical energy efficiency (η_{el}), thermal energy efficiency (η_{th}) and overall energy efficiency (η_{ov}) can be evaluated by the following equations:

$$\eta_{el} = \frac{E_{el}}{E_{Sun}} \tag{5}$$

$$\eta_{th} = \frac{E_{th}}{E_{Sun}} \tag{6}$$

$$\eta_{ov} = \frac{E_{ov}}{E_{Sun}} \tag{7}$$

E_{ov} represents the overall energetic power output, the sum of electrical (E_{el}) and thermal power output (Q_{th}).

The exergy balance equation for the tested systems can be expressed in its general form [61]:

$$\sum Ex_{in} = \sum Ex_{out} + \sum Ex_{loss} \tag{8}$$

Where Ex_{in}, Ex_{out}, and Ex_{loss} represent the input exergy rate, output exergy rate, and exergy loss.

$$Ex_{Sun} + Ex_{mass,in} = Ex_{mass,out} + Ex_{el} + \sum Ex_{loss} \tag{9}$$

Where Ex_{Sun} is sun exergy and $Ex_{\mathrm{mass,in}}$ and $Ex_{\mathrm{mass,out}}$ represents the exergy due to working fluid inlet and exists through the control volume. Ex_{el} is the electrical exergy output.

The mathematical expression for sun exergy can be given by Ref. [62,63]:

$$Ex_{Sun} = \Psi_s\, G\, A_{PV} \tag{10}$$

Where Ψ_s is the Sun factor, which calculates the input exergy received from the Sun. G is incident solar irradiation and A_{PV} is the area of PV module. T_{amb} is ambient temperature.

$$\Psi_s = \left[1 - \frac{4}{3}\left(\frac{T_{amb}}{T_{Sun}}\right)(1-\cos\phi)^{1/4} + \frac{1}{3}\left(\frac{T_{amb}}{T_{Sun}}\right)^4\right] \tag{11}$$

Where T_{Sun} is the Sun equivalent temperature assumed as a black body. Φ is the cone half-angle subtended by the Sun disc, and T_{amb} is a symbol for the ambient temperature.

The electrical exergy output (Ex_{el}) in a conventional PV module is equivalent to the electrical power output [64].

$$Ex_{el} = E_{el} \tag{12}$$

The electrical pumping power (E_{pump}) required for the pumping of fluid for thermal regulation through the flow channel can be expressed as [65]:

$$E_{Pump} = \frac{m_{wf}\, \Delta P_{drop}}{\rho_{wf}\, \eta_{pump}} \tag{13}$$

Where η_{pump} is pump operational efficiency, ΔP_{drop} is a pressure drop in the flow channel, and ρ_{wf} is the density of working fluid.

The thermal exergy power (Ex_{th}) output can be expressed as [66]:

Table 3
Density and heat capacity value viscosity of nanofluid and NePCM.

S. No	Property	Formulations	Unit	Reference
1	Density	$\rho_{nf} = \varphi\rho_{np} + (1-\varphi)\rho_{bf}$ $\rho_{NePCM} = \varphi\rho_{np} + (1-\varphi)\rho_{PCM}$	Kg/m^3	Jia et al. [69]
2	Specific heat	$c_{p,nf} = \frac{\varphi\,\rho_{np}\,c_{p,np} + (1-\varphi)\rho_{bf}\,c_{p,bf}}{\rho_{nf}}$ $c_{p,NePCM} = \frac{\varphi\,\rho_{np}\,c_{p,np} + (1-\varphi)\rho_{PCM}\,c_{p,PCM}}{\rho_{NePCM}}$	J/kg·K	[70]

$$Ex_{th} = Q_{th}\left(1 - \frac{T_{amb}}{T_m}\right) \tag{14}$$

Where T_m is mean fluid temperature and can be expressed as:

$$T_m = \frac{\left(T_{wf_2} - T_{wf_1}\right)}{ln\left(\frac{T_{wf_2}}{T_{wf_1}}\right)} \tag{15}$$

The overall exergetic power output (Ex_{ov}) is calculated by adding the electrical exergy output (Ex_{el}) and the thermal exergy output (Ex_{th}) and can be mathematically represented as:

$$Ex_{ov} = Ex_{el} + Ex_{th} \tag{16}$$

The exergy destruction (Ex_{des}) can be expressed as:

$$Ex_{des} = Ex_{Sun} - Ex_{ov} \tag{17}$$

Entropy generation (S_{gen}) is the ratio of destructive exergy and ambient temperature and can be expressed as [67]:

$$S_{gen} = \frac{Ex_{des}}{T_{amb}} \tag{18}$$

The exergy efficiencies, encompassing electrical $(\eta_{ex_{el}})$, thermal $(\eta_{ex_{th}})$, and overall efficiency $(\eta_{ex_{ov}})$, can be mathematically represented as:

$$\eta_{ex_{el}} = \frac{Ex_{el}}{Ex_{Sun}} \tag{19}$$

$$\eta_{ex_{th}} = \frac{Ex_{th}}{Ex_{Sun}} \tag{20}$$

$$\eta_{ex_{ov}} = \frac{Ex_{ov}}{Ex_{Sun}} \tag{21}$$

Equation (22) is used to evaluate the sustainability index, which measures the viability of the proposed system in terms of sustainability [68].

$$Sustainability\ index = \frac{1}{1 - \eta_{ex_{ov}}} \tag{22}$$

Table 3 presents the equations for calculating the density and specific heat of nanofluids and NePCM, where φ, ρ, and C_p represent the concentration of nanoadditive, density, and specific heat, respectively.

The relative thermal conductivity enhancement ($k_{T,enh}$) is given by the following equation. Where $k_{T,PCM}$ and $k_{T,NePCM}$ are the thermal conductivity of base PCM and NePCM, respectively.

$$k_{T,enh} = \frac{\left(k_{T,NePCM} - k_{T,PCM}\right)}{k_{T,PCM}} \times 100 \tag{23}$$

The annual electrical energy output ($E_{a,el}$) of PV and PV/T-NePCM systems is given by Ref. [71]. The variable N_s represents the total annual number of days with sunshine, while G_a represents the annual global solar irradiation.

Table 4
Economic performance parameters.

S. No.	Parameter	Expression	Reference
1	Salvage value (S)	$S = 7.5\%\ CC$	Gad et al. [72]
2	Capital recovery factor (CRF)	$CRF = \frac{R\,(1+R)^N}{(1+R)^N - 1}$	Tiwari [73]
3	Shrinkage fund factor (SFF)	$SFF = \frac{R}{(1+R)^N - 1}$	Tiwari [73]
4	Annual first cost (AFC)	$AFC = CRF \times CC$	Esfahani et al. [74]
5	Annual salvage value (ASV)	$ASV = SFF \times S$	Yousef and Hassan [75]
6	Annual maintenance cost (AMC)	$AMC = 15\%\ AFC$	Yousef and Hassan [75]
7	Total annual cost (TAC)	$TAC = AFC + AMC + ARC - ASV$	Gad et al. [72]
8	Cost of energy (COE)	$COE = \frac{TAC}{E_{en,a}}$	Yousef and Hassan [75]
9	Energy payback time (EPBT)	$EPBT = \frac{E_{emb}}{E_{en,a}}$	Tiwari and Tiwari [76]

Table 5
Environmental performance evaluation parameters.

S. No.	Parameter	Expression	Reference
1	CO_2 emissions per year	$CO_2\ emission\ per\ year = \frac{E_{emb}}{N} \times \frac{1}{(1-L_t)} \times \frac{1}{(1-L_d)} \times 0.98$	Gad et al. [72]
		$CO_2\ emission\ per\ year = \frac{E_{emb}}{N} \times 2.042$	Tiwari and Tiwari [76]
2	Net CO_2 mitigation	$Net\ CO_2\ mitigation = \left(E_{en,a} \times N - E_{emb}\right) \times 2.042$	Gad et al. [72]
3	Carbon credit gained or Enviro-economic factor	$Carbon\ credit\ gained = Net\ CO_2\ mitigation \times Cost\ per\ tons\ of\ CO_2\ mitigation$	Gad et al. [72]

$$E_{a,el} = \eta_{el,a}\,G_a\,A_{PV}\,N_S \tag{24}$$

The annual thermal energy output ($E_{a,th}$) of the PV/T-NePCM system is given by Equation 25

$$E_{a,th} = \eta_{th,a}\,G_a\,A_{PV}\,N_S \tag{25}$$

The total energy output ($E_{en,a}$) of the proposed system is the sum of the electrical and thermal energy generated on an annual basis. Table 4 displays the performance assessment parameters used in economic analysis. The system's capital cost and annual running cost are denoted as CC and ARC, respectively. R and N are the interest rate and life span. E_{emb} termed embodied energy, refers to the amount of nonrenewable energy that is used during a product's whole life cycle, from sourcing raw materials to processing, manufacturing, and transportation.

Table 5 presents the performance evaluation parameters for environmental analysis. L_t refers to the losses that occur throughout the distribution and transmission of energy, with an assumed value of 20 %. On the other hand, L_d represents the losses that occur owing to residential appliances, which is considered to be 40 %.

2.6. Uncertainty analysis

To do this experiment, the following equipment is required: an anemometer, two digital multimeters, two pressure transducers, multiple K-type thermocouples, a solar power meter, two data loggers of each 16 channel, a DC pump, and a laptop. The PV module surface temperature at several locations, including the middle and four corners, is assessed by employing K-type thermocouples. Prior to attaching the thermocouples to the surface, a calibration procedure was performed in

Table 6
Specifications of the instrument utilized in the study.

S. No.	Instrument specification	Parameters measured	Range	Make	Accuracy
1	Solar power meter	Solar radiation	0–2000 W/m^2	Tenmars TM-207	10 W/m^2
2	Anemometer	Wind speed	(0.01–100) m/s	Testo-405	0.01 m/s
3	Digital Multimeter	PV module current and voltage	Current: (0.0002–10) A Voltage: (0.2–600) V	Innova	±0.5 %
4	K-type wired thermocouple	PV upper surface temperature	(-200 -1260)°C	Nexathermal manufacturing	±0 0.4 °C
5	Pressure transducer	Coolant inlet and outlet pressure	0–0.8 Mpa	Omega	±1 %

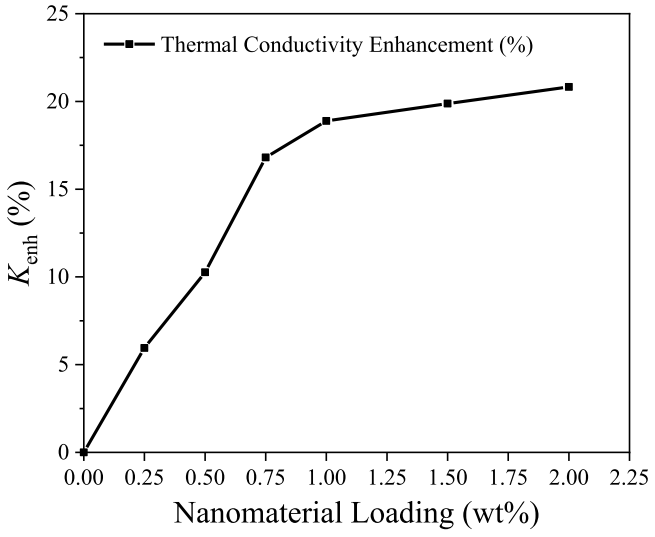


Fig. 11. Effect of nanomaterial loading on relative k_T enhancement.

a water bath to achieve a precision of 0.1 °C. The details of the instruments utilized in the current study are shown in Table 6. Using the equation provided by Al-Waeli et al. [77], the experimental cumulative error is measured to conduct the uncertainty analysis. The study's highest cumulative uncertainty is found to be less than 5 %, which shows that the fundamental parameters were accurately and reliably measured using the appropriate and calibrated instruments.

3. Results and discussions

3.1. Selection of optimum NePCM concentration and its property analysis

Fig. 11 demonstrates the enhancement in k_T when MWCNT is added to the base paraffin wax, expressed as a percentage enhancement. The graph clearly indicates that up to a concentration of 0.75 wt%, there was a significant and satisfactory rise in percentage. However, after that point, the increment became small and did not justify the increased financial expense of the nanomaterial. The low concentration of MWCNT also serves as a thermal conductive network for facilitating heat transmission within the generated NePCM. However, at more significant concentrations, nanomaterial dispersion may occur within the paraffin wax. Hence, the most suitable concentration of MWCNT for producing MWCNT-based NePCM is determined to be 0.75 wt%.

The characterization of PCM/NePCM samples is conducted using tests such as FESEM, TGA, DSC, and FT-IR. Fig. 12 displays the morphologies of MWCNT, paraffin wax, and MWCNT/paraffin wax as NePCM or nano-composite at 0.75 wt%. It is evident that paraffin possesses a sticky nature. The dispersion of MWCNTs exhibits a uniform distribution inside the paraffin matrix. Additionally, it showcases the robust integration of MWCNT and paraffin, leading to the absence of microcracks or loose surfaces. The elemental composition of the MWCNT/paraffin wax was determined using Electron Dispersion X-ray (EDX) mapping, as depicted in Fig. 13. The presence of carbon elements

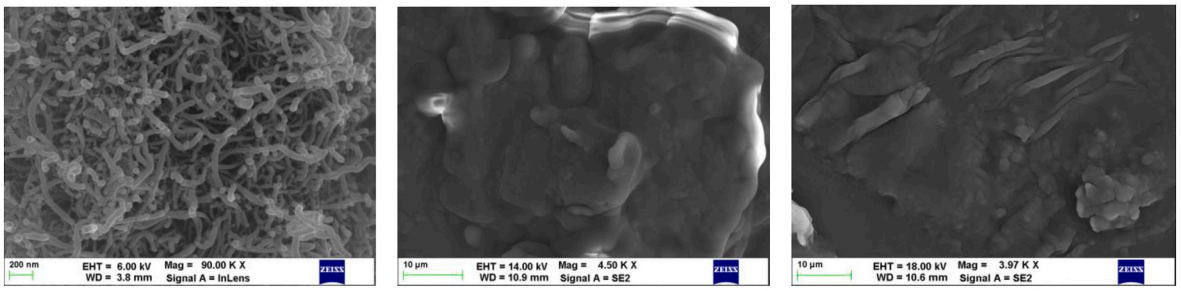


Fig. 12. FESEM images of MWCNT, paraffin wax, and MWCNT/paraffin wax as NePCM.

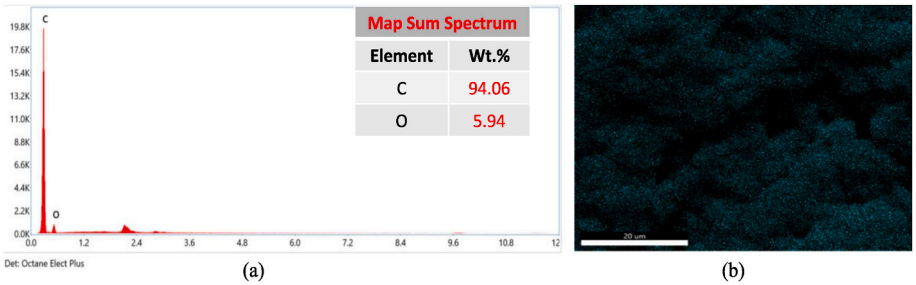


Fig. 13. (a) EDX and (b) EDS of MWCNT-enhanced paraffin wax.

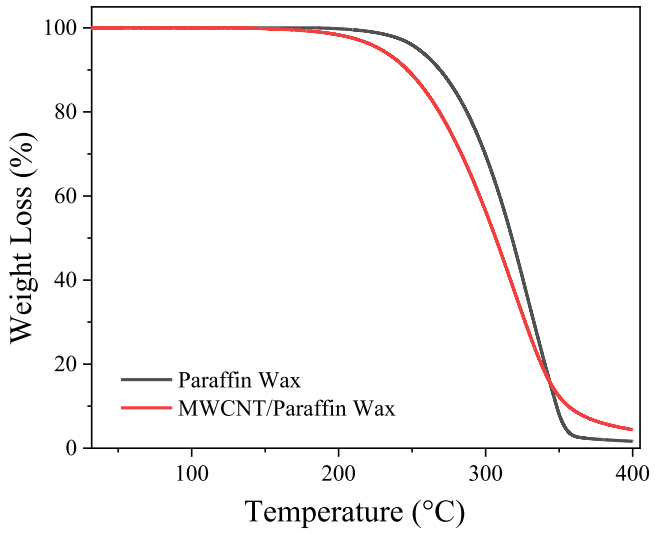


Fig. 14. TGA thermography of paraffin wax and MWCNT-enhanced PCM.

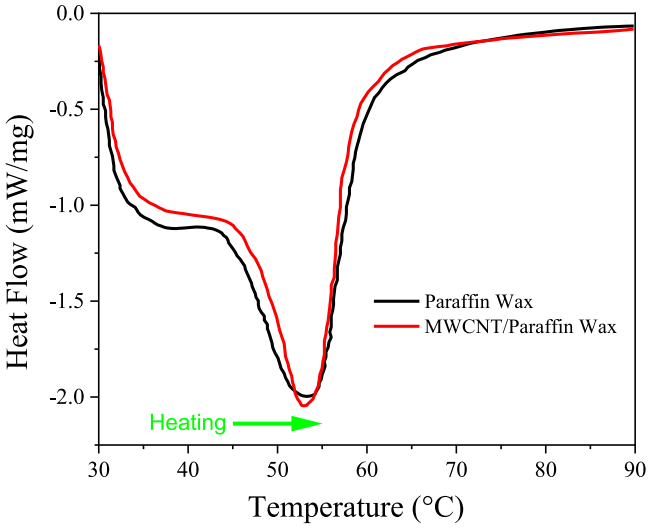


Fig. 15. DSC plots of paraffin wax and MWCNT-enhanced PCM.

is evenly spread across the surface, as observed in the Electron Dispersion Spectroscopy (EDS) spectrum. This indicates that effective stirring was employed, preventing the particles from clustering together in the paraffin matrix of selected NePCM. The carbon and oxygen weight fractions in this NePCM sample are 94.06 % and 5.94 %, respectively.

The enduring stability of base paraffin wax and MWCNT-enhanced paraffin wax as NePCM (0.75 wt%) can be assessed by analyzing the TGA pattern, as shown in Fig. 14. The stability of basic paraffin wax is evident up to 185 °C, whereas MWCNT/paraffin wax remains stable up to 165 °C since there is no apparent weight loss in the samples. However, following this temperature, the oxidation reactions resulted in noticeable weight loss. Therefore, it can be deduced that weight loss is unattainable during the process of melting due to the system's operating temperature limit being considerably lower than the critical point of 165 °C, at which the MWCNT-enhanced PCM begins to exhibit indications of weight loss.

The DSC study, depicted in Fig. 15, indicates that the phase transition occurs within a temperature range of 44.7–60.92 °C for phase change state of onset and end set. Additionally, the peak condition for the base paraffin wax during the charging process is observed at 53.45 °C. The

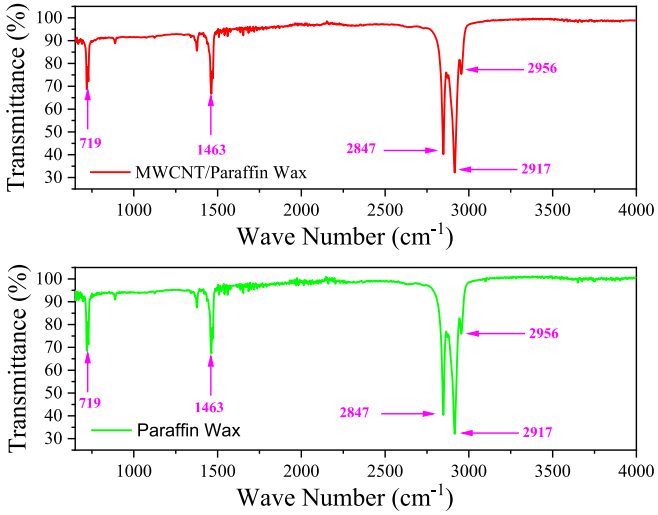


Fig. 16. Transmittance vs wave number plots for base PCM and MWCNT-enhanced PCM.

MWCNT/paraffin wax exhibits a narrow endothermic peak, indicating a quick phase transition, whereas the conventional paraffin wax has a broader temperature range for phase shift.

Fig. 16 displays the FT-IR details regarding base paraffin wax and MWCNT-enhanced paraffin wax at 0.75 wt%. Different wave numbers indicate various functional vibrations in each base and NePCM. Despite these discrepancies, the appearance of identical peaks suggests a purely physical interaction between MWCNT and paraffin wax, with no chemical changes. This demonstrates that doping MWCNTs to paraffin wax does not affect the functional group composition of the selected NePCM.

Fig. 17 presents an analysis of melting dynamics for the chosen NePCM within a finned block thermal collector. The images captured at various time intervals illustrate the evolution of phase boundaries. Initially, the NePCM existed in a solid state. Still, as the experiment unfolded, the thermal collector consistently received solar irradiation, leading to a rise in NePCM temperature and a subsequent expansion of the liquid phase region.

3.2. Ambient conditions and temperature distribution

The performance of the proposed finned multi-block thermal collector-based PV/T system is affected by several ambient circumstances, specifically solar irradiance and ambient temperature. However, unexpected changes in environmental conditions, such as cloudy weather or fluctuations in wind velocity, might introduce errors that compromise the accuracy of the outcomes. To address this issue, the investigations were carried out in September and October of 2023, when the meteorological conditions exhibited a pretty stable pattern. Fig. 18 depicts the daily variations in measured incident sun irradiance and ambient temperature between 9 a.m. and 17 p.m. The analysis demonstrates a symmetrical dispersal of solar irradiation at the time period of the experiment, resembling the shape of a bell curve. Furthermore, the peak average solar irradiation was observed to be 981.5 W/m^2 during the time frame of 12 p.m.–13 p.m.

Identifying the optimum flow rates for water and nanofluid within the system is crucial for reducing the number of tests needed. To accomplish this, PCM and DIW were used to determine the most efficient flow rate, which resulted in a significant decrease in the average temperature of the upper surface of the PV module. Fig. 19 shows a noticeable decline in the average PV module temperature with an increase in volume flow rate from 0.5 to 1.5 l/min. This reduction stems from the enhanced convective heat transfer coefficient between the PV

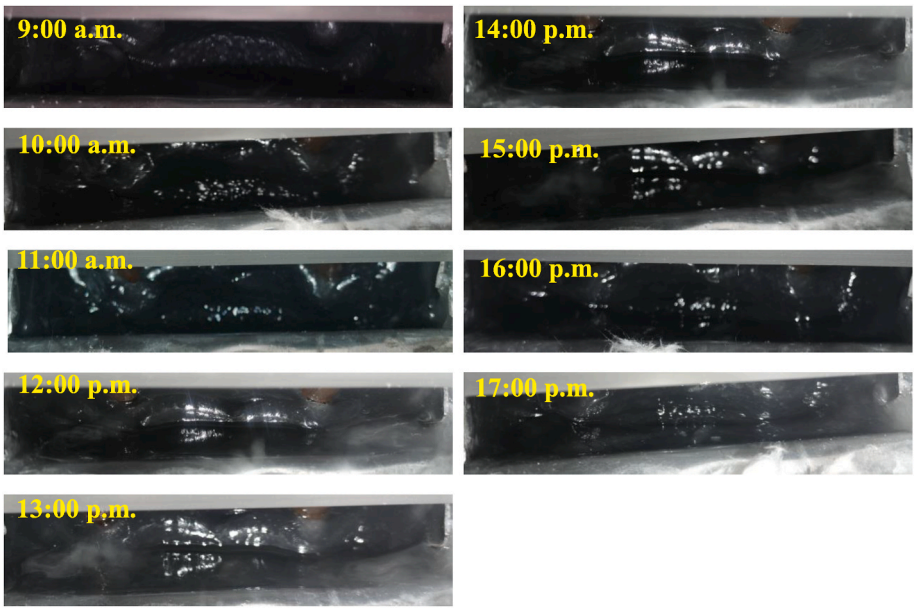


Fig. 17. Photographs of melting behavior of the chosen NePCM.

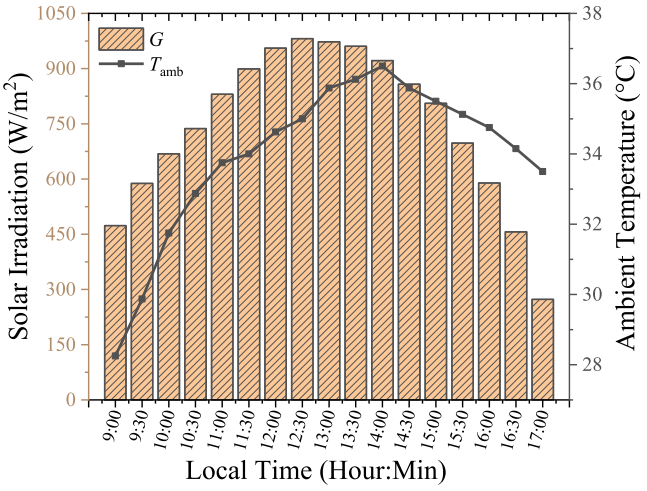


Fig. 18. Plot for average solar irradiation and ambient temperature vs time of day.

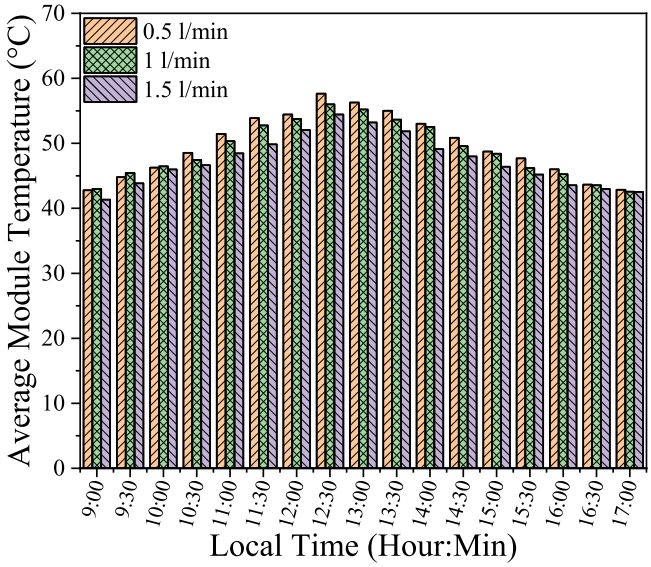


Fig. 19. Effect of water flow rates on average PV module temperature.

module and the circulating working fluid as the volume flow rate of water increases, facilitating more effective heat dissipation from the PV module's surface. This higher flow rate enables the PV module to operate at a lower temperature. While it's possible to further increase the coolant flow rate, this comes with drawbacks, such as increased pumping power requirements and the onset of vibrations in the tested system at higher flow rates, limiting experimentation within the range of 0.5–1.5 l/min. Consequently, the performance evaluation of the proposed PV/T-NePCM system with DIW and other nanofluids (0.25, 0.5, and 0.75 wt%) at a flow rate of 1.5 l/min is presented.

Fig. 20 depicts temperature profiles, displaying the average temperatures of PV modules for both conventional PV and the investigated pioneering PV/T-NePCM system utilizing different water and nanofluid-based coolants, including pure DIW, MWCNT/DIW-0.25, MWCNT/DIW-0.5, and MWCNT/DIW-0.75. The results showed that these systems achieved reductions in average PV module temperature of 13.35 °C, 14.54 °C, 15.45 °C, and 16.44 °C, respectively, compared to the conventional PV module surface temperature. The PV/T-NePCM systems exhibit a measurable percentage drop in both average and maximum PV module temperature ranging from 22.8 % to 28.1 % and 24.1 %–29.7 %, respectively, when compared to the traditional PV module. The MWCNT/DIW coolant at a concentration of 0.75 % demonstrates a more significant decrease in the surface temperature of the PV module. This can be attributed to the combined impact of two heat absorption mediums, NePCM and nanofluid, simultaneously. This integration increases the ability of the nanofluid to transmit heat, therefore enhancing the capacity of the working fluid to carry heat. Table 7 presents the reduction in PV module temperature for investigated PV/T-NePCM systems with different coolants.

Fig. 21 presents a comparative analysis of temperature contour diagrams between a conventional PV module and a proposed PV/T-NePCM system incorporating MWCNT/DIW-0/75. Images from an infrared camera for thermography were captured at various time intervals to enable the comparison. The observed trends in solar irradiation and upper surface temperatures across different time intervals exhibit similar patterns. As solar irradiation increases until noon, the

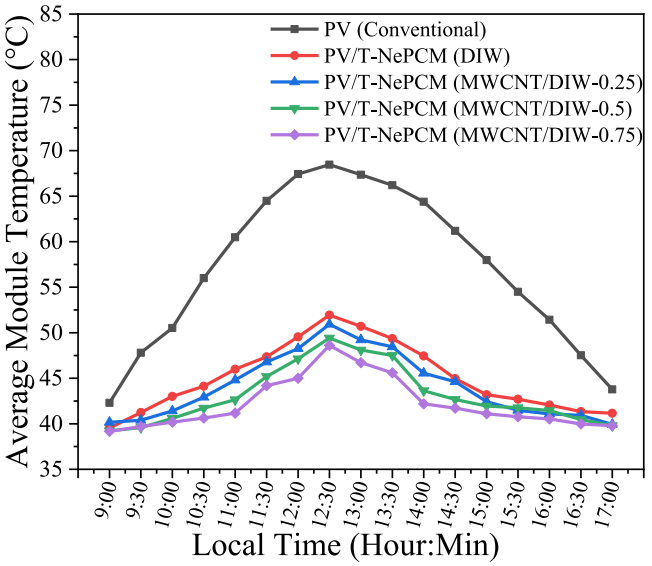


Fig. 20. Average PV module temperature vs local time of day.

Table 7
Reduction in PV module temperature for investigated PV/T-NePCM system with different heat transfer fluids.

Investigated heat transfer fluids	Reduction in upper surface module temperature of PV/T-NePCM systems compared to conventional PV module (°C)	
	Average	Maximum
DIW	13.39	16.51
MWCNT/DIW-0.25	14.54	18.53
MWCNT/DIW-0.5	15.45	19.04
MWCNT/DIW-0.75	16.44	20.34

upper surface temperature of the PV modules rises. On the other hand, a decrease in the average PV surface temperature correlates to a reduction in solar irradiation in the afternoon. Notably, the irradiance distribution is more concentrated in the middle section of the PV modules, leading to localized temperature increases compared to other areas. The proposed PV/T-NePCM system demonstrates superior thermal regulation by effectively reducing the upper PV module temperatures. This outcome underscores the enhanced cooling performance achieved by integrating NePCM and nanofluid in the PV/T system, highlighting the system's capability for effective temperature control and regulation.

3.3. Energy analysis

Fig. 22 displays a comparison of average E_{el} and Q_{th} between a conventional PV module and a PV/T-NePCM system using various coolants. The study's results reveal that the average thermal energy output (Q_{th}) for each examined scenario of PV/T-NePCM systems is 159.87 W, 165.86 W, 174.88 W, and 184.53 W for coolants pure DIW, MWCNT/DIW-0.25, MWCNT/DIW-0.5, and MWCNT/DIW-0.75 correspondingly. Concerning pure DIW, the PV/T-NePCM systems employing nanofluid-based coolants showed a respective increase in Q_{th} of 3.75 %, 9.39 %, and 15.43 % for MWCNT/DIW-0.25, MWCNT/DIW-0.5, and MWCNT/DIW-0.75 coolants. Adding MWCNT into the DIW results in an increase in the k_T of the coolant. As a result, the suggested PV/T-NePCM systems experience an increase in Q_{th} when MWCNT/DIW is used as a working fluid. The measured percentage rise in average E_{el} for the examined DIW and nanofluid-based coolants was determined to be 17.99 %, 18.7 %, 19.3 %, and 19.97 %, respectively, in comparison to the absence of a coolant system. The application of MWCNT/DIW nanofluid can potentially reduce temperature fluctuations in PV modules. The inclusion of nanomaterial in the nanofluid boosts its ability to regulate heat, leading to better control of PV module surface temperature. This is accomplished by minimizing the heat resistance between the PV module and the coolant, significantly enhancing the total system performance.

Fig. 23 illustrates the variations in η_{el} over the local time of day for conventional PV module and suggested PV/T-NePCM systems. These systems use different coolants, such as pure DIW and nanofluid-based coolants with varying concentrations of MWCNT, namely MWCNT/DIW-0.25, MWCNT/DIW-0.5, and MWCNT/DIW-0.75. The recorded average η_{el} for the examined systems is 12.09 %, 13.18 %, 13.26 %, 13.32 %, and 13.38 %, respectively. The decrease in η_{el} caused by an increase in the temperature of solar cells is not as noticeable in the PV/T-NePCM system as in conventional uncooled PV module. The phenomenon occurs because NePCM, with its increased k_T, has an innate ability to efficiently control the rise in cell temperature. This is accomplished through their ability to absorb a significant amount of heat during successive cycles of melting and freezing.

Fig. 24 depicts a comparative analysis of η_{el}, η_{th}, and η_{ov} for each instance under examination. These cases include the traditional PV module and PV/T-NePCM systems that use various coolants. The average η_{el} is reported to increase by 9.01 %, 9.68 %, 10.17 %, and 10.67 % for pure DIW and MWCNT/DIW coolants, respectively, compared to the system without a coolant. The temperature differential between coolant inlet and outlet temperatures is more significant in nanofluid and NePCM integrated systems than in pure DIW coolant systems. Therefore, the average η_{th} of the examined coolants was determined to be 6.28 %, 9.4 %, and 10.89 % compared to the pure DIW coolant. Hence, the comprehensive assessment of PV/T-NePCM systems elucidates their overall system efficiency. Table 8 thoroughly summarizes the energy performance metrics for both traditional PV and PV/T-NePCM systems.

3.4. Exergy analysis

Fig. 25 depicts the comparison of the average Ex_{el}, Ex_{th}, and Ex_{ov} for a PV/T-NePCM system using various coolants, such as pure DIW, MWCNT/DIW-0.25, MWCNT/DIW-0.5, and MWCNT/DIW-0.75. The Ex_{th} for the studied coolants is 3.95 W, 4.2 W, 4.38 W, and 4.5 W, respectively and MWCNT/DIW-0.75 coolant demonstrates the highest enhancement of 13.92 % in comparison to the pure DIW coolant.

Fig. 26 depicts a comparative examination of the average $\eta_{ex_{el}}$, $\eta_{ex_{th}}$, and $\eta_{ex_{ov}}$ for several coolant instances that were evaluated. The $\eta_{ex_{el}}$ of the tested coolants were found to be 12.77 %, 12.84 %, 12.9 %, and 12.97 % for DIW, MWCNT/DIW-0.25, MWCNT/DIW-0.5, and MWCNT/DIW-0.75 coolants, respectively. The $\eta_{ex_{el}}$ of the PV/T-NePCM system was observed to be enhanced by 10.18 %, 10.79 %, 11.3 %, and 11.91 % when employing DIW and MWCNT/DIW-nanofluid as coolants, compared to conventional uncooled system. The $\eta_{ex_{th}}$ for the examined coolants are measured to be 1.59 %, 1.69 %, 1.76 %, and 1.81 % respectively. The PV/T-NePCM systems using MWCNT/DIW nanofluid show a significant increase in $\eta_{ex_{th}}$, ranging from 6.29 % to 13.84 %, compared to DIW-based coolant systems. The decrease in exergy efficiency for electricity, as opposed to energy efficiency for electricity, can be ascribed to the incorporation of energy quality in exergy analysis. This term incorporates not only the amount of energy but also its accessibility for carrying out vital tasks. Energy losses due to irreversibility are detected in the studied PV/T-NePCM systems that use DIW and MWCNT/DIW at varying concentrations as the heat transfer medium. These irreversibilities refer to the dissipation of thermal energy, inefficiencies in heat transfer, and losses in converting solar irradiation into electricity. The existence of irreversibility in the system being studied causes a reduction in the thermodynamic excellence of the energy, resulting in a lower $\eta_{ex_{el}}$ relative to the η_{el}. The present study

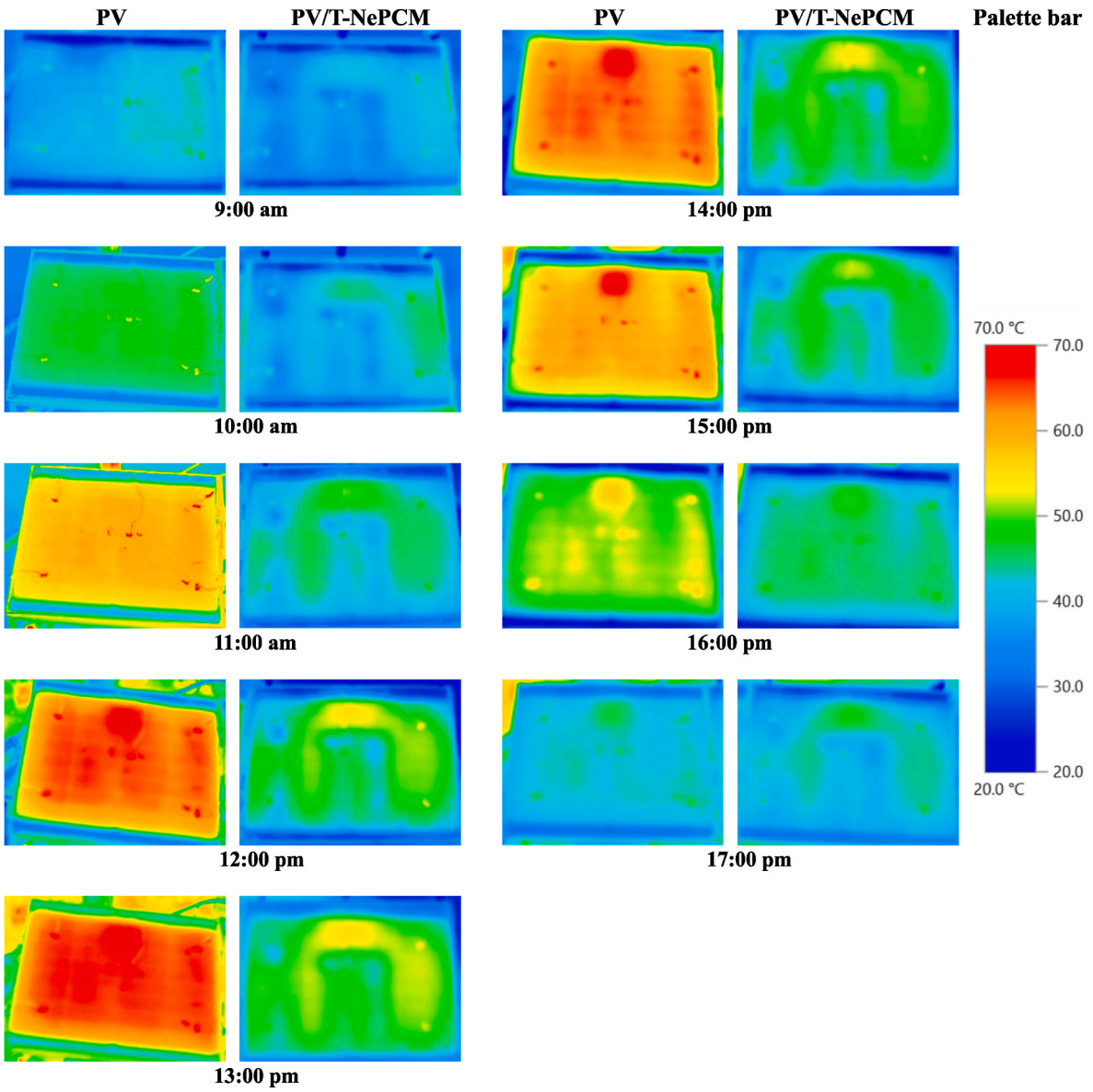


Fig. 21. Comparison of temperature contour diagrams.

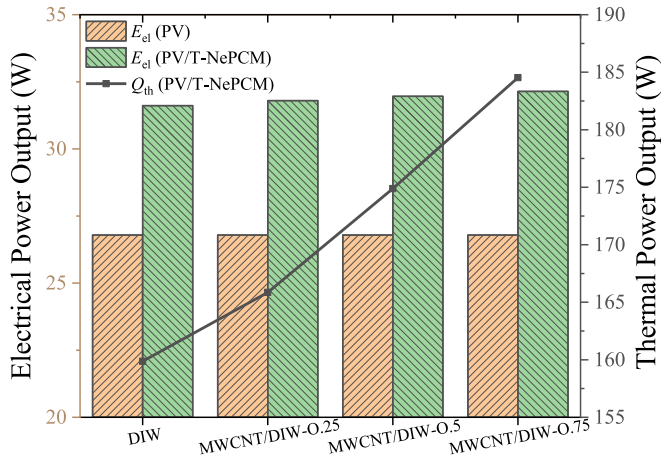


Fig. 22. Average E_{el} and Q_{th} for investigated systems with different coolants.

examined the improvement of both $\eta_{ex_{el}}$ and $\eta_{ex_{th}}$ in PV/T-NePCM systems.

Fig. 27 depicts the changes in Ex_{el}, Ex_{th}, Ex_{ov}, and Ex_{des} time for the PV/T-NePCM with MWCNT/DIW-0.75 coolant. Ex_{th} is somewhat smaller than Ex_{el} since electricity is a superior kind of energy. Therefore, the primary determinant responsible for the majority of exergy is assigned to electrical exergy. The exergy destruction reaches its maximum value at noon because of the PV module's peak electrical power generation at the time of higher solar irradiation, together with the notably elevated temperature of the investigated PV/T-NePCM system. The temperature difference between the investigated system and its surroundings could lead to the dissipation of exergy. The Ex_{des} at around 12:30 p.m. reached a value of 279.81 W during the period of highest sun irradiation.

S_{gen} is a relevant parameter for illustrating the irreversibility of a system. The goal is to reduce the generation of entropy in the system under study to enhance its efficiency. Fig. 28 illustrates a comparison between the average S_{gen} and exergy destruction of a PV/T-NePCM system using different coolants: DIW, MWCNT/DIW-0.25, MWCNT/DIW-0.5, and MWCNT/DIW-0.75. The results of the study suggest that

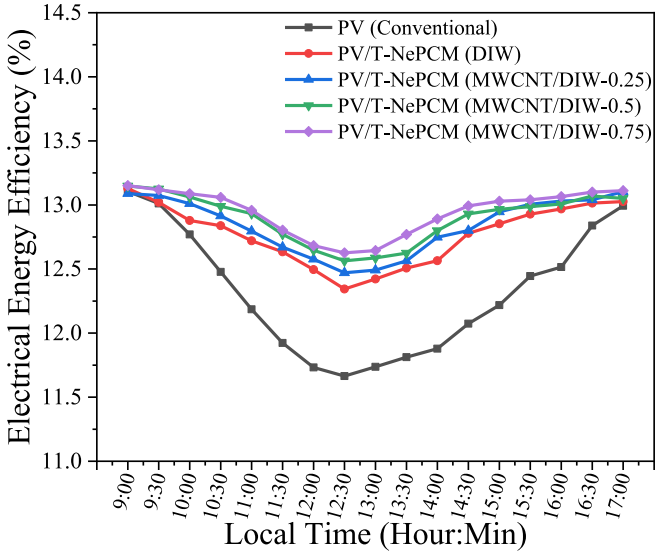


Fig. 23. η_{el} vs local time of day.

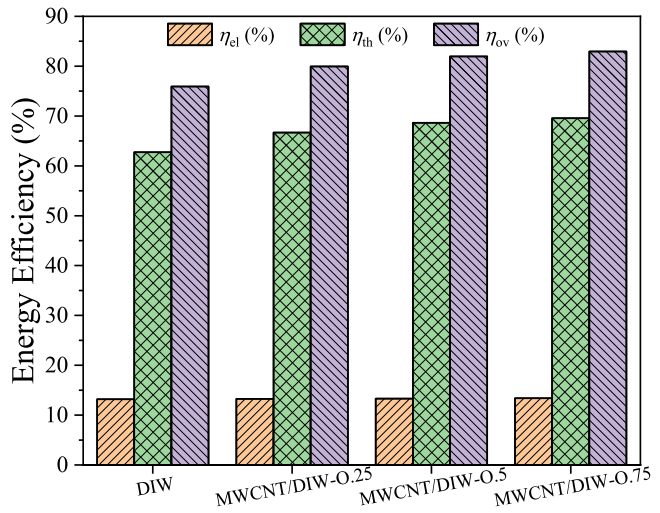


Fig. 24. Average η_{el}, η_{th}, and η_{ov} of PV/T-NePCM system for different coolants.

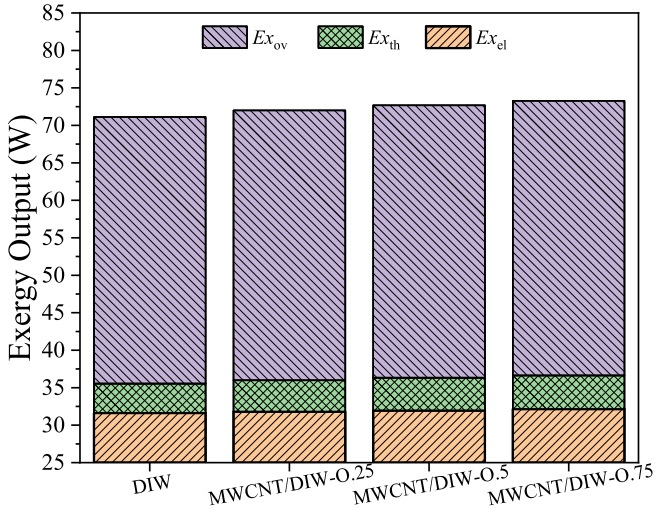


Fig. 25. Average Ex_{el}, Ex_{th}, and Ex_{ov} for PV/T-NePCM system utilizing different coolants.

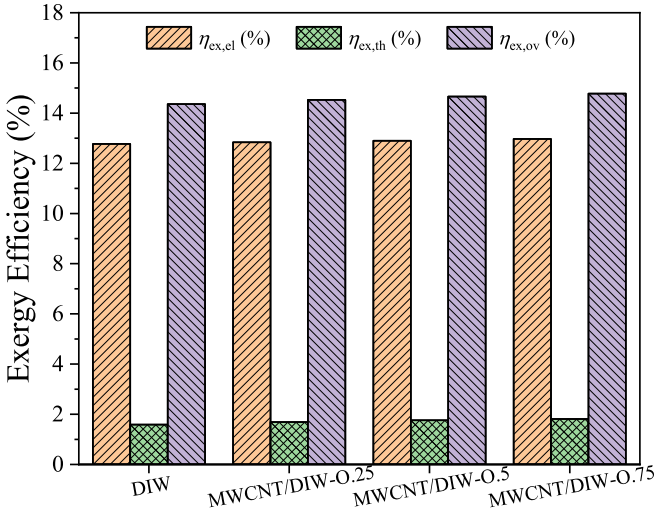


Fig. 26. Comparison of $\eta_{ex_{el}}$, $\eta_{ex_{th}}$, and $\eta_{ex_{ov}}$ of PV/T-NePCM system utilizing different coolants.

Table 8
Summary of energy performance matrices of conventional PV and PV/T-NePCM systems.

Energy performance matrices	PV	PV/T-NePCM			
		DIW	MWCNT/DIW-0.25	MWCNT/DIW-0.5	MWCNT/DIW-0.75
E_{el} (W)	26.79	31.61	31.8	31.96	32.14
Q_{th} (W)	–	159.87	165.86	174.88	184.53
E_{ov} (W)	26.79	191.48	197.66	206.84	216.67
η_{el} (%)	12.09	13.18	13.26	13.32	13.38
η_{th} (%)	–	62.74	66.68	68.64	69.57
η_{ov} (%)	12.09	75.92	79.94	81.96	82.95

the average S_{gen} of the tested DIW and different MWCNT/DIW nanofluid-based coolants decreases by 3.95 %, 4.15 %, 4.31 %, and 4.44 %, respectively, compared to without a cooling system. Similarly, a decrease in the average exergy destruction is seen. Hence, incorporating active and passive cooling methods in a PV/T system significantly reduces entropy generation. The reduction of exergy destruction is achieved by combining these strategies, which enhance heat transfer by utilizing nanofluid to transfer heat from the PV upper surface to the coolant running through channel configuration. The suggested system design aims to reduce the amount of exergy destruction by improving the even dispersal of heat transmission and reducing temperature differences by using separate finned NePCM blocks. Table 9 offers a detailed summary of the exergy metrics for both conventional PV and PV/T-NePCM systems.

Fig. 29 displays the sustainability index of the PV/T-NePCM system using DIW and various nanofluid-based coolants. This term is contingent upon the system's $\eta_{ex_{ov}}$ and calculated by Equation (22) and is employed to characterize the performance of the proposed system in terms of sustainability. The MWCNT/DIW-0.75 coolant-based PV/T-NePCM system has a sustainability index of 1.1734, making it the most sustainable system in the study.

3.5. Economic analysis

An economic analysis has been conducted to assess the financial feasibility of the PV/T-NePCM system in the present investigation. The annual average global radiation in India is recorded as 5.5 kWh per unit

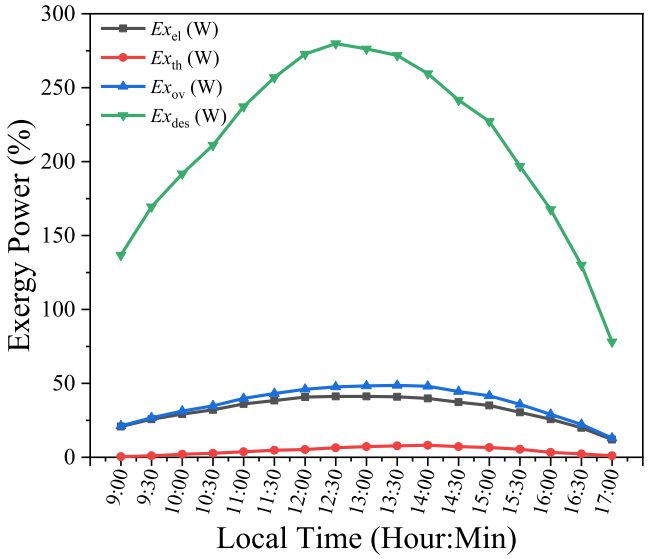


Fig. 27. Ex_{el}, Ex_{th}, Ex_{ov}, and Ex_{des} vs local time for proposed PV/T-NePCM with MWCNT/DIW-0.75 coolant.

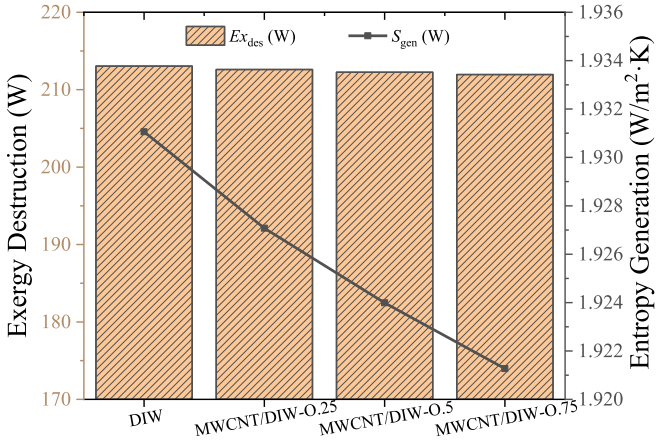


Fig. 28. S_{gen} and Ex_{des} for PV/T-NePCM system under different coolants.

Table 9
Summary of exergy performance matrices of conventional PV and PV/T-NePCM systems.

Energy performance matrices	PV	PV/T-NePCM			
		DIW	MWCNT/DIW-0.25	MWCNT/DIW-0.5	MWCNT/DIW-0.75
Ex_{Sun} (W)	248.6	248.6	248.6	248.6	248.6
Ex_{el} (W)	26.79	31.61	31.8	31.96	32.14
Ex_{th} (W)	–	3.95	4.2	4.38	4.5
Ex_{ov} (W)	26.79	35.56	36	36.34	36.64
Ex_{des} (W)	221.81	213.04	212.6	212.26	211.96
$\eta_{ex_{el}}$ (%)	11.59	12.77	12.84	12.9	12.97
$\eta_{ex_{th}}$ (%)	–	1.59	1.69	1.76	1.81
$\eta_{ex_{ov}}$ (%)	11.59	14.36	14.53	14.66	14.78

area per day, based on an assumption of 300 days of sunshine per year. The capital cost breakdown of the investigated PV/T-NePCM (0.75 %) system utilizing nanofluids is presented in Table 10. The anticipated average rate of exchange between the USD and the Indian Rupee (INR) for the year 2023 is projected to be 82.1931. The standard 50 W

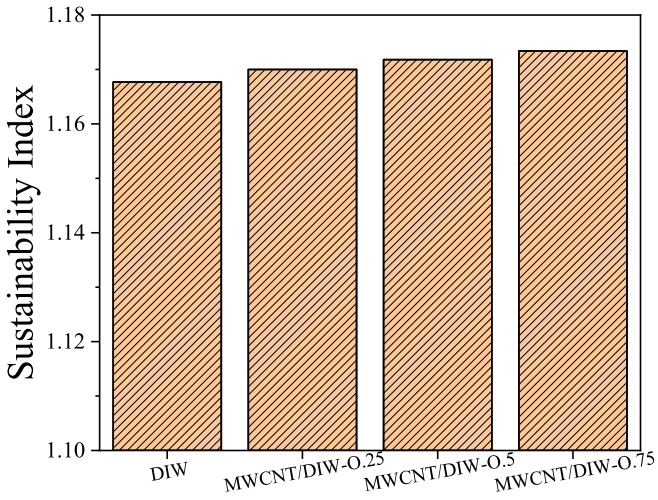


Fig. 29. Sustainability index for PV/T-NePCM system under different coolants.

Table 10
Capital cost breakdown of nanofluid-based PV/T-NePCM system.

S. No.	System component	Description	Cost per item (INR)	Quantity	Total cost of item (INR)
1	PV	Polycrystalline-50 W	2500	1 No.	2500
2	Framework	Composed of Iron	1000	1 No.	1000
3	Thermal collector	Finned absorber plate-based thermal regulating blocks	500	6 Nos.	2500
		Cu pipe for fluid circulation	1000		1100
4	Thermal conductive paste	To minimize thermal resistance	300	2 Nos.	600
5	Insulation	Glass wool and plywood sheet	800		800
6	PCM	Paraffin wax	215	7 kg	1505
7	Nanoadditive	MWCNT	300	10 g	3000
8	Pneumatic pipe	For fluid circulation	695		695
9	Pump	For coolant pumping	1100	1 No.	1400
10	Storage tank	For coolant storage	300	1 No.	500
11	Heat exchanger		5000	1 No.	5000
12	Flow control valve	For changing the flow rate	100	2 Nos.	200
13	Installation cost				1500
Total capital cost (CC)					22300 INR (271.31 USD)

polycrystalline PV system has a capital cost of USD 42.58, including the price of a framework, while the total retail capital cost of the suggested system is USD 271.31. The power tariff for commercial consumers in India is assumed to be 8 INR/kWh (0.097 USD/kWh) and the mean price of natural gas is 5.5 INR/kWh (0.068 USD/kWh).

Table 11 displays the computation of economic performance indicators for the assessed systems using a rate of interest of 8 % and an average lifespan of 20 years. The cost of energy (COE) for the conventional uncooled system and the proposed nanofluid-cooled PV/T NePCM system is determined to be 5.64 INR (0.069 USD) and 5.24 INR (0.063 USD), respectively. The energy payback time, which measures the time required to recover the energy invested in a system, is found to be 4.93

Table 11
Computation of economic performance indicators for the assessed systems.

Tested Systems	CC (INR)	N (Year)	R (%)	S (INR)	AFC (INR)	ASV (INR)	AMC (INR)	TAC (INR)	$E_{en,a}$ (kWh/year)	COE (INR/kWh)
PV	3500	20	8	262.5	356.48	5.74	53.47	404.22	71.63	5.64
PV/T-NePCM	22300	20	8	1672.5	2171.3	36.55	340.7	2575.45	491.49	5.24

Table 12
E_{emb} of proposed nanofluid-based PV/T-NePCM system.

S. No.	Component material	Quantity (kg)	Energy density (kWh/kg)	E_{emb} (kWh)
1	PV	0.54 × 0.665 m^2	910 kWh/m^2	326.78
2	Framework material-Iron	3	8.89	26.67
3	PCM	7	11.67	81.69
4	Nanoadditive	0.065	68.55	4.46
5	Flow configuration material-Cu	0.85	19.61	16.67
6	Thermal collector material-Al	1.5	55.28	82.92
7	Utilized plastic material	5	19.44	97.2
8	Adhesives	0.01	88.96	0.89
Total embodies energy				637.28

years for PV systems and 1.29 years for the PV/T-NePCM system.

3.6. Environmental analysis

Table 12 illustrates the embodied energy (E_{emb}) associated with manufacturing the proposed PV/T system, integrated with NePCM and nanofluid. Due to its increased material requirements, this PV/T-NePCM system has a higher total E_{emb} than the conventional uncooled system. The E_{emb} of all materials comprising the PV/T-NePCM system was determined to be 637.28 kWh. The predominant contributor to the total E_{emb} of the proposed system was the PV module, constituting 51.28 % of the overall energy. Despite the PCM holding significance in the total system weight, its contribution to the total E_{emb} was relatively lower. This is attributed to the cost-effectiveness of obtaining the PCM compared to other system components. Notably, the nanoadditive exhibited a relatively higher energy density of 68.55 kWh/kg, allowing for its utilization in lesser concentrations to improve the heat transfer properties of nanofluids and NePCM through k_T augmentation.

Table 13 represents the calculation for environmental performance evaluation parameters such as CO_2 emission, net CO_2 mitigation, and Enviro-economic factor/Carbon credits gained from conventional PV and proposed PV/T-NePCM system with a life span of 20 years. The projected cost for mitigating each tonne of CO_2 is USD 10, based on the international price of carbon. The proposed nanofluid-cooled PV/T-NePCM system outperformed the conventional uncooled system in terms of the significant amount of net CO_2 mitigated and the resulting Carbon credits gained.

4. Concluding remarks

This study introduces an innovative system design that combines both active and passive cooling methods in PV/T systems with concurrent nanoadditive integration in both heat transfer fluid and PCMs. The system comprises a thermal collector featuring a multi-block finned absorber plate type thermal collector. The primary objective is to enhance the uniformity of heat transfer distribution and decrease temperature stratification within each multi-block containing NePCM. Therefore, this study investigated the performance assessment of innovative PV/T-NePCM system utilizing DIW and nanofluid-cooled heat transfer fluids, namely MWCNT/DIW-0.25, MWCNT/DIW-0.5 and MWCNT/DIW-0.75 in the real Indian climate conditions, considering energy, exergy, economic, and environmental aspect. The study's conclusion remarks are as follows.

- The PV/T-NePCM system, utilizing MWCNT/DIW nanofluid as the heat transfer fluid at an optimal concentration of 0.75 wt%, demonstrates an energy efficiency and exergy efficiency superior by 9.26 % and 2.93 %, respectively, compared to the system employing DIW as the base fluid.
- The tested DIW and MWCNT/DIW nanofluid-based coolants at 0.25, 0.5, and 0.75 wt% exhibit a notable reduction in average entropy generation by 3.95 %, 4.15 %, 4.31 %, and 4.44 % respectively, when compared to a scenario without a cooling system.
- The PV/T-NePCM system utilizing MWCNT/DIW nanofluid at a concentration of 0.75 wt% demonstrates a remarkable sustainability index of 1.1734, establishing its position as the most environmentally sustainable system among all the tested alternatives.
- The economic evaluation demonstrates that the PV/T-NePCM system exhibits a cost of energy amounting to 5.24 INR/kWh (0.063 USD/kWh), in contrast to the conventional PV module, which registers a higher price at 5.64 INR/kWh (0.069 USD/kWh). These calculations were conducted with an 8 % interest rate and assumed a system lifespan of 20 years.
- Regarding the context of environmental sustainability, the calculated total embodied energy of the proposed nanofluid-cooled PV/T-NePCM system amounts to 637.28 kWh. With a projected system lifespan of 20 years, the associated annual CO_2 emissions were found to be 0.0651 tonnes, while the net CO_2 mitigation over the system's lifespan is estimated at 18.77 tonnes.

The proposed nanofluid-cooled multi-block finned absorber plate-based thermal collector exhibited superior performance in terms of energetic, exergetic, economic, and environmental (4E) perspectives. However, the present investigation requires further exploration in the following areas.

- The effect of hybrid nanofluid flow through the proposed multi-block finned NePCM-based thermal collector can be analyzed.
- An evaluation of the proposed system's performance can be carried out utilizing corrugated flow channels.
- Explored the proposed design by incorporating hybrid NePCM and optimizing the thickness of PCM block containers.

Table 13
Environmental analysis of conventional uncooled PV and proposed nanofluid-cooled PV/T-NePCM system.

Tested Systems	E_{emb} (kWh)	Life (N)	$E_{en,a}$ (kWh/year)	International Carbon price (USD/tonne of CO_2)	CO_2 emission (kg/year)	Net CO_2 mitigation (kg)	Carbon credits gained (USD)
PV	353.45	20	71.63	10	36.09	2204.03	22.04
PV/T-NePCM	637.28	20	491.49	10	65.07	18771.13	187.71

CRediT authorship contribution statement

Praveen Kumar Tyagi: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rajan Kumar:** Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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